Master in Business Analytics & Data Science
Research Capstone in Operations Research

The Price of Sophistication: When Do Spatial and Robust Modeling Pay in Carbon-Aware Data-Center Scheduling?

A Day-Ahead Migration Scheduler and a Complexity–Value Decision Rule

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AI Use Declaration. Generative AI tools (Anthropic Claude) were used as a research and engineering assistant under the author's direction: for code implementation support, experiment orchestration, literature triage, and drafting assistance. All research questions, modelling decisions, experimental designs, and interpretations are the author's, developed with the supervisor; all results were produced by version-controlled, pre-registered code reviewed by the author, and all text was reviewed and edited by the author.

Abstract

Data-center operators can shift compute toward cleaner hours and regions, and recent work applies distributionally robust optimization (DRO) to carbon-aware scheduling. The unexamined question is which modelling layers actually pay for their complexity. This thesis builds a working day-ahead carbon-aware migration scheduler and characterises the value of each layer. Over the honest baseline (carbon-aware scheduling with no inter-region transfer, $\Phi = 0$, on the same feasible set), active migration cuts daily $\mathrm{CVaR}_{0.95}$ emissions by 4.0–9.9% (Western 4.0%, Eastern 9.9%, Diversified 9.0%), and this deterministic transfer channel is the dominant lever; the carbon-computing literature independently finds spatial shifting dominates temporal shifting. We then isolate the value of *passive* sophistication. Using a Mahalanobis–Wasserstein DRO solved as a second-order cone program, a pre-registered shuffled-marginals falsification test across three real US/Canada grids spanning the dependence spectrum (the US West, an Ontario-anchored Eastern belt, and an engineered solar/wind/hydro portfolio), with an Iberia–France anchor, returns a replicated **null**: encoding spatial covariance adds below 0.4% of CVaR, surviving Ledoit–Wolf shrinkage, residualization, Benjamini–Hochberg correction, walk-forward validation, and a per-cell equivalence test. A mean-ablation shows the covariance signal is genuine but masked by the mean field (a mean-dominance bound), and a tail-dependence analysis shows the residual dependence is non-elliptical and upper-tail independent, hence invisible to covariance-based ambiguity sets; a one-paragraph copula appendix confirms even the maximal coupling recovers nothing. Finally we ask when robustness itself pays. Following the price of robustness of Bertsimas and Sim, a DRO over day-ahead forecast error buys a small worst-day (CVaR) tail reduction at a mean premium, with a value of the stochastic solution near zero, and pays over deterministic transfer only past an emergency-severity crossover $M^{\star} \approx 3$; tested against real data-grounded worst-tail emergencies across 17 zones, that crossover never activates. The thesis contributes a working scheduler with honest savings, a screening rule for when passive dependence modelling fails, and a price-of-robustness decision rule with a tested, data-grounded bound, all under 194 unit tests and pre-registration.

Contents

1 Introduction

1.1 Context

Data centers consumed an estimated 1–2% of global electricity and their demand is growing rapidly with AI workloads [8]. Unlike most industrial loads, compute is unusually *flexible*: many batch and training workloads can be delayed by hours or moved between sites with little loss of utility. This movement is *logical*, jobs relocate over the network between an operator's data centers, and does not require the regions' power grids to be physically interconnected; spatial correlation of carbon intensity therefore matters as information for the scheduler, not as an electrical constraint. Because the carbon intensity of grid electricity varies strongly by hour and by region, driven by wind, solar, hydro availability, and demand, this flexibility lets a data-center operator act as a grid-aware participant, deferring work from dirty hours to clean ones [18, 21]. Google's carbon-intelligent computing platform demonstrated such temporal shifting at production scale [18], and recent systems work extends it to variable-capacity operation [22].

Scheduling against *tomorrow's* carbon intensity, however, means scheduling under uncertainty. Distributionally robust optimization (DRO) addresses this by optimizing against the worst distribution in an ambiguity ball around the empirical distribution [6], and has been applied to carbon-aware scheduling with carbon intensity as the uncertain quantity [7]. Existing formulations treat each region's carbon intensity essentially in isolation. Yet carbon intensities of electrically interconnected regions co-move: weather fronts traverse neighbouring balancing areas, and shared solar fleets create synchronized clean periods. It is therefore natural to model carbon intensity as a stochastic *vector* with a full spatial covariance, the extension studied here.

1.2 Problem statement and research questions

Each modelling layer, spatial covariance, distributional robustness, and richer copulas, adds estimation and computational cost, and is worthwhile only if it buys measurably better schedules. The literature has assumed, rather than measured, that this sophistication pays. This thesis builds the scheduler and measures the value of each layer directly.

RQ1. How much can a day-ahead carbon-aware migration scheduler cut daily $\mathrm{CVaR}_{0.95}$ emissions relative to carbon-aware scheduling without inter-region transfer, and which lever drives the saving?

RQ2. When does adding passive modelling complexity (spatial covariance, or a non-elliptical copula) improve out-of-sample tail emissions, and at what cost?

RQ3. When does robustifying against day-ahead forecast error pay over deterministic transfer, and do real grids ever reach that regime? What decision rule follows?

1.3 Contributions

The thesis makes four contributions. (i) *A working day-ahead scheduler with honest savings*: active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the carbon-aware no-transfer baseline ($\Phi = 0$, same feasible set) across three real grids, with the transfer lever dominant. (ii) *A screening rule*: a pre-registered shuffled-marginals falsification test and mean-ablation diagnostic isolate the value of spatial covariance, yielding a replicated null (gaps below 0.4% of CVaR, robust to estimator choice, multiple-testing correction, and walk-forward), with a mean-dominance bound (Proposition 1) as the a-priori certificate of when passive dependence modelling cannot help. (iii) *A price-of-robustness decision rule*: framing robustness as a complexity–value trade-off after Bertsimas and Sim, the DRO over day-ahead forecast error buys tail reduction at a mean premium and pays only past an emergency-severity crossover $M^{\star} \approx 3$ that real data-grounded emergencies across 17 zones never reach. (iv) *A reproducible pipeline*: version-controlled, CI-tested (194 unit tests), pre-registered, with archived license-safe summary tables for every reported number.

1.4 Thesis structure

Section 2 reviews the carbon-aware scheduling and DRO literature and locates the gap: the value of modelling sophistication has been assumed, not measured. Section 3 develops the day-ahead migration scheduler, the DRO model, the falsification test, and the evaluation protocol. Section 4 reports the scheduler savings and the value of each modelling layer: where the value comes from, the screening rule for passive dependence, and the price-of-robustness decision rule with the severity crossover. Section 5 interprets the findings and states limitations. Section 6 concludes with the practitioner decision rule and future work.

2 Literature Review

2.1 Carbon-aware computing

Carbon-aware scheduling exploits temporal and spatial variation in grid carbon intensity. In production at Google, Radovanović et al. [18] describe a system which computes day-ahead "virtual capacity curves" that throttle flexible compute during high-carbon hours; measured impact on latency-sensitive workloads is negligible while load shifts measurably toward cleaner hours. Wiesner et al. [21] analyse the potential of temporal shifting across regions and workloads, finding the benefit depends strongly on the grid's marginal mix and the workload's flexibility. Wijayawardana and Chien [22] study cloud-VM scheduling on datacenters whose capacity varies with grid conditions (including carbon-coupled capacity, where a constant carbon budget forces capacity to fall as intensity rises) and document the goodput cost of capacity variation for long-running jobs. This systems literature establishes operational levers (deferral, throttling, capacity coupling) that we adopt as constraints, but it treats future carbon intensity as a point forecast; uncertainty is handled implicitly, by re-planning.

2.2 Distributionally robust optimization and the price of robustness

DRO optimizes against the worst-case distribution within an ambiguity set, and the Wasserstein ball around the empirical distribution has emerged as the canonical data-driven choice, with tractable finite-dimensional reformulations [6, 16]. For costs linear in the uncertain vector, type-2 Wasserstein DRO with a Mahalanobis ground metric reduces to a mean term plus a covariance-weighted regularizer, an interpretable "mean plus $\varepsilon \sqrt{x^\top \Sigma x}$" objective solvable as a second-order cone program (SOCP) [2]. Hall et al. [7] apply Wasserstein DRO to carbon-aware data-center scheduling with virtual capacity curves, treating each region's carbon intensity as the uncertain quantity and demonstrating robustness gains over sample-average approximation. The risk functional we target, CVaR_β, is standard for tail-focused evaluation [19]. Beyond second moments, recent work robustifies the *dependence structure* itself: Fan et al. [4] place both the marginals and the *copula* within a Wasserstein ball, the natural home for the non-elliptical structure our copula schedulers target.

Robustness is never free. Bertsimas and Sim [3] framed this as the *price of robustness*: protecting against worst-case realizations trades a guaranteed degradation in nominal performance for a bounded improvement in the tail, and the operator chooses where on that frontier to sit. This lens is central to our RQ3: we do not ask whether robustification *can* help (it can, in the tail) but whether the price it charges is paid back under conditions real grids actually reach. The mean-premium-for-tail-reduction trade-off we report for the day-ahead scheduler is a direct instance of their frontier.

2.3 Modelling cross-region dependence

Cross-region dependence enters a Mahalanobis–Wasserstein model only through the off-diagonal blocks of the covariance matrix. Estimating large covariance matrices from short panels is noisy; shrinkage estimators [17] are the standard remedy and serve here as a robustness arm. Beyond second moments, dependence is fully described by the copula [9, 20]; tail-dependence coefficients quantify joint extremes that correlation cannot see, and the Gaussian copula has zero asymptotic tail dependence for any correlation below one [5], a property we use as a diagnostic benchmark.

Vine (pair-copula) constructions make high-dimensional non-elliptical dependence tractable [9, 10, 12], with established sequential structure-selection procedures [11], and copula-based scenario generation for stochastic programs is well developed [13]. Whether richer non-elliptical structure (vine copulas, or copula-ambiguity DRO [4]) improves a *stochastic program* depends on whether it changes the optimal decision out-of-sample; we test this directly and report it in Appendix B.

2.4 The gap: value has been assumed, not measured

The carbon-aware scheduling literature establishes that *temporal* shifting (deferral to clean hours) delivers measurable savings, confirmed operationally at Google and in recent studies [18, 21]. The natural extension is *spatial* shifting (migration toward clean regions), and the distributed-computing literature finds spatial shifting dominates temporal shifting in potential [21], independently motivating the transfer channel studied here. Yet the value of *modelling* that spatial structure (via spatial covariance, copulas, or distributional robustness) has been *assumed*, not measured. The DRO literature applies sophisticated ambiguity sets to uncertain vectors without isolating the marginal value of the cross-coordinate dependence they encode, and neither literature offers a controlled mechanism for *why* spatial structure should or should not matter. This thesis fills that gap: we build a working day-ahead scheduler that actively migrates compute and measure the savings (RQ1), isolate the value of passive modelling sophistication via a falsification test with a causal mechanism (RQ2), and frame robustification as a price-of-robustness trade-off, testing when that price is paid in real data (RQ3).

3 Methodology

3.1 Research design overview

The design tests whether passive (covariance-based) spatial modelling of carbon intensity adds value to robust day-ahead scheduling, and measures the value of active transfer separately. We hold the optimization machinery fixed and manipulate exactly one object (the cross-region blocks of the covariance supplied to the scheduler) then measure the consequence on held-out tail emissions. Three primary region sets spanning the dependence spectrum (with two earlier panels corroborating) serve as replications; three nested constraint regimes guard against the result being an artifact of a particular feasible-set geometry; a robustness battery (estimators, multiple-testing correction, walk-forward) guards against statistical artifacts; and two further experiments (mean-ablation; tail dependence) identify the mechanism. Throughout, a pre-registration discipline removes analyst degrees of freedom.

3.2 The scheduling problem

Compute is scheduled across R regions over a $T = 24$-hour day. The decision is $x \in \mathbb{R}_{\geq 0}^{R \times T}$, where $x_{r,t}$ is compute power (MW) in region r at hour t. Carbon intensity $\rho_{r,t}$ (gCO$_2$eq/kWh) is uncertain; realized emissions are linear, $\langle \rho, x \rangle = \sum_{r,t} \rho_{r,t} x_{r,t}$.

3.3 Distributionally robust formulation: day-ahead forecast-error hedging

The robust layer hedges *day-ahead forecast error*. Each day the scheduler commits against a day-ahead point forecast $\bar{\rho}$ of the carbon field (the hour-of-day climatological mean on the training window, updated daily); the realized field departs by a residual $\xi = \rho - \bar{\rho}$ whose empirical distribution $\hat{\mathbb{P}}$, pooled over 2021–2024, defines the ambiguity. We minimize worst-case expected emissions over a type-2 Wasserstein ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius ε centred on that residual distribution, with Mahalanobis ground metric induced by the empirical residual covariance $\hat{\Sigma} \in \mathbb{R}^{RT \times RT}$:

$$\min_{x \in \mathcal{X}} \sup_{\mathbb{Q} \in \mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x \rangle\big] \;=\; \min_{x \in \mathcal{X}} \langle \bar{\rho}, x \rangle + \varepsilon \sqrt{x^\top \hat{\Sigma}\, x}, \tag{1}$$

by Wasserstein duality for linear costs [2, 6], where $\bar{\rho}$ is the empirical mean field. With the Cholesky factorization $\hat{\Sigma} = LL^\top$ the penalty becomes $\varepsilon \|L^\top \operatorname{vec}(x)\|_2$. Introducing an epigraph variable η (distinct from the hour index t), (1) is the second-order cone program

$$\min_{x \in \mathcal{X},\, \eta \geq 0} \langle \bar{\rho}, x \rangle + \varepsilon\, \eta \quad \text{s.t.} \quad \left\| L^\top \operatorname{vec}(x) \right\|_2 \leq \eta, \tag{2}$$

solved to global optimality (CLARABEL). The radius ε interpolates from the deterministic LP ($\varepsilon = 0$) to increasingly conservative schedules.

Where spatial correlation enters. Partitioning $\hat{\Sigma}$ into $T \times T$ blocks Σ_{rs} by region pair, the quadratic form splits as

$$x^\top \hat{\Sigma} x = \sum_r x_r^\top \Sigma_{rr} x_r + \sum_{r \neq s} x_r^\top \Sigma_{rs} x_s, \tag{3}$$

so the off-diagonal (spatial) blocks are the *only* channel by which cross-region dependence can alter the schedule. The experiment manipulates exactly these blocks.

3.4 A ladder of modelling layers

The thesis studies schedulers arranged by sophistication: **(0)** carbon-blind; **(1)** per-region deterministic (each region by its own mean forecast); **(2)** per-region robust (single-region Wasserstein DRO, no cross-region structure); **(3)** joint covariance (the SOCP (1) with the full spatial Σ, the passive spatial extension); and **(4)** joint copula (non-elliptical dependence; Appendix B). The screening test (Section 3.7) measures the marginal value of (3) over (2). Active transfer ($\Phi > 0$, inter-region flows) is orthogonal to this passive hierarchy and is the dominant lever, studied in Sections 4.2 and 4.9.

3.5 Feasible set and constraint regimes

The feasible set $\mathcal{X}$ collects the linear constraints below; introducing an auxiliary flexible-load variable $x^{\mathrm{f}} \in \mathbb{R}^{R\times T}_{\geq 0}$, the scheduler solves (1) over

$$0 \leq x_{r,t} \leq \bar{x}_{r,t}, \qquad \forall r,t \qquad \text{(C0: capacity)} \tag{4a}$$

$$x_{r,t} = \alpha_r W_r\, p_{r,t} + x^{\mathrm{f}}_{r,t}, \qquad \forall r,t \qquad \text{(C2a: flex/inflex split)} \tag{4b}$$

$$\textstyle\sum_t x^{\mathrm{f}}_{r,t} = (1-\alpha_r)\, W_r, \qquad \forall r \qquad \text{(C2b: work conservation)} \tag{4c}$$

$$|x_{r,t} - x_{r,t-1}| \leq \Delta_r, \qquad \forall r,t \qquad \text{(C3: ramp)} \tag{4d}$$

$$\textstyle\sum_{t\in\mathcal{W}} x^{\mathrm{f}}_{r,t} \geq \gamma\,(1-\alpha_r)\, W_r, \qquad \forall r \qquad \text{(3a: deferral deadline)} \tag{4e}$$

$$\pi_{r,t}\, x_{r,t} \leq \bar{P}, \qquad \forall r,t \qquad \text{(3b: thermal/PUE)} \tag{4f}$$

$$\textstyle\sum_{r,t} \bar{\rho}_{r,t}\, x_{r,t} \leq B \qquad \text{(3d: carbon budget)} \tag{4g}$$

where W_r is region r's daily work (utilization fixed at 80%, $W_r = 960$ MWh); $\alpha_r \in \{0.30, 0.50, 0.75\}$ is the inflexible fraction following the fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); $\Delta_r = 15$ MW/h is the ramp limit; $\mathcal{W} = [0,7]$ and $\gamma = 0.20$ set the deferral deadline; and $\pi_{r,t} = \mathrm{PUE}_0 + \kappa \max(\mathcal{T}_{r,t} - \mathcal{T}_{\mathrm{set}}, 0)$ is the temperature-coupled power-usage multiplier (data, hence a constant per cell), capped at effective power $\bar{P}$. Every constraint is linear, so $\mathcal{X}$ is polyhedral and the program (1) remains an SOCP. (We follow the source labelling: C0–C3 are the base operational constraints and the **3**· series is adapted from the SoCC'25 formulation [22]; a further aggregate hourly cap $\sum_r x_{r,t} \leq p_{\max}$ is implemented but inactive in the reported regimes.)

The capacity ceiling is constant ($\bar{x}_{r,t} = 50$ MW) except under constraint **3c**, the carbon-coupled variable capacity adapted from Wijayawardana and Chien [22], which replaces it with

$$\bar{x}_{r,t} = x_{\min} + (x_{\max} - x_{\min})\, \mathrm{CFE}_{r,t}/100, \tag{5}$$

where $\mathrm{CFE}_{r,t}$ is the training-mean carbon-free-energy share, capacity falls as the grid gets dirtier. The bounds $(x_{\min}, x_{\max}) = (50, 75)$ were calibrated on training data to a loosely binding ("Goldilocks") regime: the constraint binds on 28–37% of region-hours, leaving positive slack and capacity margin, so it shapes the schedule without freezing it.

Three nested regimes report results as a function of operational tightness: **R3** (reference) imposes C0 + C2 + C3; **R1** (lean) adds the deferral deadline 3a; and **R2** adds the thermal ceiling 3b (climate cases) or the variable-capacity ceiling 3c (interconnection cases). The carbon budget 3d is available but inactive in the reported regimes. Each regime is solved as a *complete* model with all of its constraints active jointly; the nesting probes the null's robustness to operational tightness, and is not a test of any single constraint's effect in isolation.

3.6 Model validation on a small instance

Before running the falsification experiment we check that the optimization model itself behaves correctly. The full model is validated *as a whole*, with all constraints of (4) active simultaneously, on a small, human-checkable *two-region* instance (two regions, eight hours, 10 MWh per-cell capacity, $W = 30$ MWh each), rather than each constraint in isolation: correctness is a property of the constraints' *interaction*, and a set that every constraint satisfies alone can still be jointly binding in unexpected ways. We use two regions specifically so the *cross-region* interaction is exercised: a shared aggregate hourly cap $\sum_r x_{r,t} \leq P_{\text{agg}}$ couples the regions, which must then compete for the same clean hours. We fix an analyst-anticipated outcome, perturb one input at a time, and verify the solved schedule moves the anticipated way (Figure 1). At baseline the two regions' clean windows overlap, and the aggregate cap forces them to split the contested hours (panel A). Making region 1's clean window dirty drives it to cede those hours to region 2 and spread to its own next-cleanest hours, with each region's work still conserved (panel B). Tightening the shared cap removes joint headroom, so both regions spread further in time rather than co-locating in the clean window (panel C). All three responses match the hand-anticipated solution and conserve work per region to machine precision. This model-level check is complementary to, and distinct from, the software test suite (Section 3.11, which pins component behaviour such as feasible-set equivalence) and the falsification experiment below (which probes a scientific hypothesis, not model correctness).

3.7 The falsification test: shuffled marginals

The scheduler is fitted twice, with covariances differing in exactly one respect:

$$\left(\hat{\Sigma}^{\text{shuf}}\right)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}, & r \neq s, \end{cases} \tag{6}$$

preserving every region's marginal distribution and within-region autocorrelation while destroying only cross-region dependence. Each fitted schedule x is evaluated by the out-of-sample conditional value-at-risk of daily emissions $e_i = \langle \rho_i, x \rangle$ realized on test day i; CVaR is a coherent risk measure

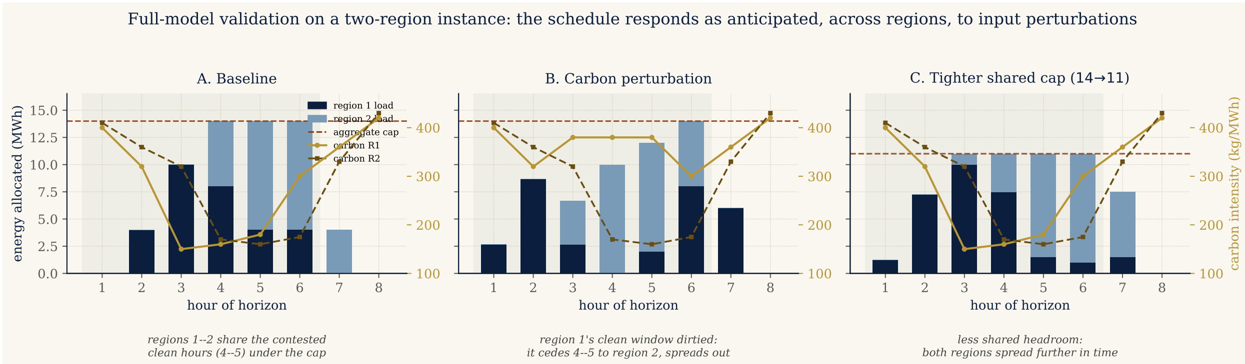

Figure 1: Full-model validation on a two-region instance. The complete feasible set (capacity, work conservation, ramp, deferral deadline) plus a shared aggregate hourly cap is solved jointly; stacked bars show how the two regions share each hour, the dashed line is the aggregate cap, the curves are each region's carbon field, the green band the deadline window. **A:** baseline, the regions split the contested clean hours (4–5) under the cap. **B:** after dirtying region 1's clean window, it cedes those hours to region 2 and spreads out. **C:** after tightening the shared cap, both regions spread further in time. Each response is the one an analyst anticipates, confirming the model encodes the intended cross-region behaviour.

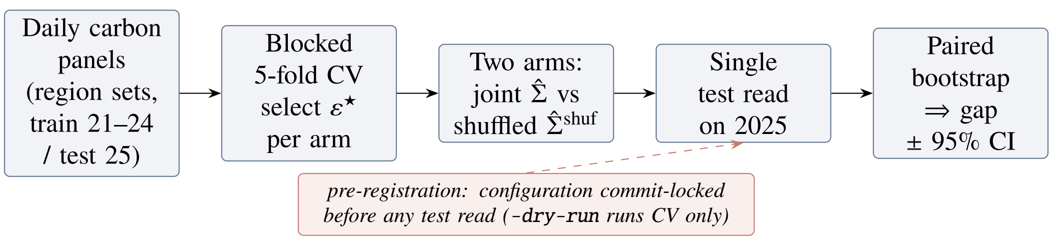

Figure 2: The pre-registered falsification pipeline. Cross-validation and schedule fitting touch only the training years; the 2025 test set is read once, after the configuration is commit-locked, and the spatial gap is read off a paired day-bootstrap.

[14] standard in stochastic programming [15]. We use the Rockafellar–Uryasev form [19]

$$\mathrm{CVaR}_\beta(e) = \min_{\tau\in\mathbb{R}} \left\{\tau + \tfrac{1}{1-\beta}\,\mathbb{E}\big[(e-\tau)_+\big]\right\}, \qquad \beta = 0.95, \tag{7}$$

estimated as the empirical mean of the worst 5% of test days. The *spatial gap* is the difference between the two arms,

$$\mathrm{gap} = \mathrm{CVaR}_{0.95}\big(e^{\mathrm{shuf}}\big) - \mathrm{CVaR}_{0.95}\big(e^{\mathrm{joint}}\big), \tag{8}$$

positive when the joint covariance helps. The test is a screening tool: built to detect a spatial benefit if one exists, it isolates the *value* of passive covariance modelling from the *validity* of spatial correlation in the data, and validates the mean-dominance bound (Proposition 1). Figure 2 sketches the pipeline and Algorithm 1 gives the full protocol.

Algorithm 1 Shuffled-marginals experiment (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); regimes {R3, R1, R2}; α-levels $\{0.30, 0.50, 0.75\}$; grid $\mathcal{E} = \{0, 0.1, 1, 10, 100, 1000\}$
for each regime, each α, each arm $\Sigma \in \{\text{joint}, \text{shuf}\}$ **do**
  for each $\varepsilon \in \mathcal{E}$ **do** ▷ blocked 5-fold time-series CV
    for each contiguous fold k **do**
      fit $\bar{\rho}, L$ on training folds; solve SOCP (1); record validation $\mathrm{CVaR}_{0.95}$
  $\varepsilon^{\star} \leftarrow \arg\min_{\varepsilon}$ mean validation CVaR ▷ per arm, independently
  refit on full training set at $\varepsilon^{\star}$; evaluate *once* on 2025 ▷ single test read
paired bootstrap (1000 resamples, fixed seed) on test days $\Rightarrow$ 95% CI for the gap (8)
Pre-registration: configuration commit-locked before any test read; `-dry-run` gate runs CV only

3.8 Data

Carbon intensity is hourly lifecycle intensity from Electricity Maps (academic licence), 2021–2025, 43,824 hourly observations per zone; panels are built on a common local clock per region set, trained on 2021–2024 and tested once on 2025. Three primary US/Canada region sets span the dependence spectrum: **(1)** the *US West*, California, Nevada and Phoenix (CISO, BANC, LDWP, NEVP, AZPS); **(2)** an *Ontario–Eastern belt* (CA-ON, NYISO, MISO, PJM); and **(3)** an *engineered heterogeneous portfolio* (CISO (solar), ERCOT (wind), BPAT (hydro), spanning *different* interconnections) deliberately chosen so clean periods do *not* align: the adversarial best case for diversification. This spread is deliberate on both ends of the spectrum. The two correlated, within-interconnection grids are the *best case for the spatial hypothesis*, the most cross-region correlation a covariance model could possibly exploit, and double as a stress test rather than a realistic deployment (no operator clusters data centers within one interconnection). A realistic, globally dispersed fleet looks instead like the heterogeneous portfolio, low and mixed correlation, so the practically relevant case is the one where diversification *should* help most. The null holding at both ends is therefore stronger than a null on either alone. Two earlier panels corroborate: an *Iberia–France* European set (ES, PT, FR), where residual correlation collapses to the diurnal cycle, and the California/Nevada subset of the US West before Phoenix was added at the supervisor's request. Temperature for constraint 3b is hourly ERA5 2 m air temperature (Open-Meteo), one load-weighted point per zone. The licensed raw data is never committed; derived summary tables are archived in the repository for traceability.

3.9 Validation, diagnostics, and the mean-dominance bound

Correlation validation (RQ1). Raw Pearson correlation conflates co-movement with a shared diurnal cycle. We therefore report correlations of *residuals* after removing each zone's hour-of-day mean (local clock), with overlay plots of standardized intensity for summer and winter weeks.

Robustness battery. Every experiment is re-run with (i) Ledoit–Wolf shrinkage covariance [17]; (ii) covariance estimated from seasonal (monthly-mean) residuals; (iii) covariance from AR(1) residuals. A pre-registered agreement rule counts a spatial effect only if positive and sign-stable across both residual estimators. Bootstrap p-values across all cells receive Benjamini–Hochberg correction at $q = 0.05$ [1]; a walk-forward refit (train 2021–2023, test 2024) checks temporal stability; a log-denser ε grid checks grid sensitivity.

Mean-ablation (RQ3, causal). To test whether the mean field masks the covariance, the mean used for *scheduling* is transformed while evaluation stays on real emissions: LEVEL equalizes regional time-averages (keeping hour-of-day shape); FLAT sets a constant mean, making the mean term constant under fixed demand, a covariance-only world in which any joint-vs-shuf difference is attributable purely to the spatial blocks.

Tail dependence (RQ3, structural). For each region pair we estimate the upper and lower tail-dependence coefficients $\chi_{\mathrm{U}}(q) = \Pr(U > q, V > q)/(1 - q)$ and $\chi_{\mathrm{L}}(p) = \Pr(U < p, V < p)/p$ on rank pseudo-observations of residual intensity, comparing χ_{U} at $q = 0.95$ with the Gaussian-copula value at the matched correlation. Since the Gaussian copula, the dependence family a covariance ball can represent, is asymptotically tail-independent [5], a positive empirical excess would be exploitable tail structure invisible to the DRO; and since elliptical copulas force $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, radial asymmetry diagnoses non-elliptical dependence.

3.10 Phase 2: richer dependence models (summary)

The covariance ball represents only *elliptical* dependence. To foreclose the "wrong object" objection, Appendix B re-runs the falsification with Gaussian, lower-tail Clayton, and the maximal (comonotone) copula, fitted to the empirical rank structure on the same feasible set. The result confirms the screening rule and is summarized in one paragraph there; Proposition 1 below explains why it must hold.

Why the dependence model is second-order. Two facts make the dependence model second-order to the mean field. The first is a decomposition: by the translation invariance of CVaR, writing the carbon field as mean plus zero-mean residual $\rho = \bar{\rho} + \xi$ and noting $\langle \bar{\rho}, x \rangle$ is deterministic,

$$\mathrm{CVaR}_\beta\big(\langle \rho, x \rangle\big) = \underbrace{\langle \bar{\rho}, x \rangle}_{\text{dependence-independent}} + \mathrm{CVaR}_\beta\big(\langle \xi, x \rangle\big), \tag{9}$$

so for a *fixed* schedule the dependence model moves only the residual term. The second is an a-priori bound on how far it can move the *optimal* value.

Proposition 1 (Spatial gap control, SOCP)**.** *Let* $\mathrm{OPT}(\Sigma)$ *be the optimal value of the SOCP* (2) *for a covariance* Σ*, and let* Σ^{shuf} *be its block-diagonal (cross-region-zeroed) counterpart with* $\Sigma^{\mathrm{off}} = \Sigma - \Sigma^{\mathrm{shuf}}$*. Then the spatial gap obeys*

$$\left|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\right| \;\le\; \varepsilon \left(\max_{x\in\mathcal{X}} \|x\|_2\right) \left\|\Sigma^{\mathrm{off}}\right\|_2^{1/2}, \tag{10}$$

which vanishes as $\varepsilon \to 0$ *(no robustification) or* $\Sigma^{\mathrm{off}} \to 0$ *(no cross-region structure).*

Proof. The two programs share the feasible set $\mathcal{X}$ and differ only in the conic penalty. For fixed x, using $|\sqrt{a} - \sqrt{b}| \le \sqrt{|a-b|}$ and the Rayleigh bound $|x^\top \Sigma^{\mathrm{off}} x| \le \|\Sigma^{\mathrm{off}}\|_2 \|x\|_2^2$,

$$\left|\varepsilon\sqrt{x^\top \Sigma x} - \varepsilon\sqrt{x^\top \Sigma^{\mathrm{shuf}} x}\right| \le \varepsilon\sqrt{|x^\top \Sigma^{\mathrm{off}} x|} \le \varepsilon \|x\|_2 \|\Sigma^{\mathrm{off}}\|_2^{1/2}.$$

Since the infimum is 1-Lipschitz in the sup-norm of the objective, two objectives that differ pointwise by at most δ over a common feasible set have optimal values differing by at most δ. The feasible set $\mathcal{X}$ is a bounded polytope (the capacity bounds C0), hence compact, so $\max_{x\in\mathcal{X}} \|x\|_2$ is attained; taking that maximum gives (10). □

The bound is a conservative *certificate*, not a tight prediction. Evaluated on each grid's training covariance (with $\varepsilon^\star = 1$, the a-priori feasible bound $\max_x \|x\|_2 = \bar{x}\sqrt{R\,T\,u}$ where $\bar{x}$ is the per-cell cap and u the utilization, and $\|\Sigma^{\mathrm{off}}\|_2$ the spectral norm of the destroyed cross-region blocks), its right-hand side is 7.4% (Eastern US–Canada), 9.6% (Diversified) and 12.7% (Western US) of $\mathrm{CVaR}_{0.95}$; the realized gaps (Table 1) never exceed 0.23%, two orders of magnitude smaller. The theorem therefore caps the spatial gap at the few-percent scale *before any experiment is run*, and the experiments then pin it far below that cap. What the bound buys is not tightness but structure: the spatial gap is governed by ε and the cross-region block norm, and is forced to zero in the no-robustification and no-cross-structure limits. Most of the remaining slack lives in the feasible-diameter factor $\max_x \|x\|_2$, an a-priori worst case; substituting the realized optimal $\|x^\star\|_2$ would tighten the certificate considerably, at the cost of making it post-hoc rather than a-priori. We keep the a-priori form deliberately, since its purpose is to bound the gap *before* the experiment, not to predict it. The *tight* ceiling is measured, not bounded: the comonotone arm of Section 4.8 realizes the supremum over copulas, and across scenario seeds its gain is centred on zero with $|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$, at the sampling-noise floor. The mean-ablation of Section 4.6 bounds the same quantity from a second, independent angle. Both put Λ an order of magnitude below the schedule's mean-driven value; hence the mean field pins the schedule and the dependence model, elliptical or not, is immaterial. The prediction that even Clayton cannot recover material value is confirmed in Section 4.8.

Computational considerations. Each instance is small: the decision vector has $RT \leq 5 \times 24 = 120$ entries and the feasible set adds $O(RT)$ linear constraints, so the SOCP (2) solves in 0.19–0.26 s and the CVaR-SAA LP (11), which appends $S = 1000$ epigraph variables and constraints, in 0.9–1.3 s (median over five repetitions, CLARABEL/HiGHS, commodity laptop). The cost is in the *number* of instances, not their size: one region-set experiment solves 3 regimes $\times$ 3 $\alpha \times$ 2 arms $\times$ (6 $\varepsilon \times$ 5 folds + 1 test read) = 558 SOCPs, and the full Phase 1 battery (region sets $\times$ {sample, Ledoit–Wolf, seasonal, AR(1), two ablations, walk-forward, fine grid}) is on the order of 10^4 SOCP solves; Phase 2 adds $3 \times 3 \times 3 \times 4 = 108$ CVaR-SAA LPs. Total wall-clock for the entire study is under two hours on one core, so reproduction is cheap.

3.11 Reproducibility

All code is version-controlled (private repository, continuous-integration test suite, 194 unit tests, including a feasible-set equivalence test pinning the Phase 2 CVaR scheduler to the Phase 1 constraint set); experiment configurations are commit-locked before test-set evaluation; bootstrap and scenario seeds are fixed; and every number reported in Sections 4–5 traces to an archived summary table. Solvers: CLARABEL/ECOS for the SOCP; HiGHS for the CVaR-SAA linear programs and deterministic LP baselines.

4 Results

4.1 The spatial assumption is valid (RQ1)

The premise holds. After removing each zone's hour-of-day mean, cross-region correlations of carbon intensity in the two electrically coupled US/Canada cases are strong and survive de-seasonalization: in the US West, CISO–NEVP 0.78 and CISO–LDWP 0.72 (shared solar resource and weather across WECC); in the Ontario–Eastern belt, CA-ON–NYISO 0.73 and MISO–PJM 0.70 (weather fronts traversing the Eastern Interconnection). The engineered portfolio is the low, *heterogeneous* end, not a zero-correlation grid: its residual pairs span 0.00 (ERCOT–BPAT), 0.29 (CISO–ERCOT) and 0.37 (CISO–BPAT), mean $\bar{r} \approx 0.2$, a deliberately mixed generation profile. Such partly-low, heterogeneous correlation is *diversifiable* (it drives the largest mean-ablation signal, Section 4.6), unlike the US West's strong but *uniform*, common-mode correlation. Figure 3 shows the residual correlation matrices; overlay plots of standardized intensity (archived with the code) confirm visible co-movement in summer and winter weeks for the coupled cases. Together

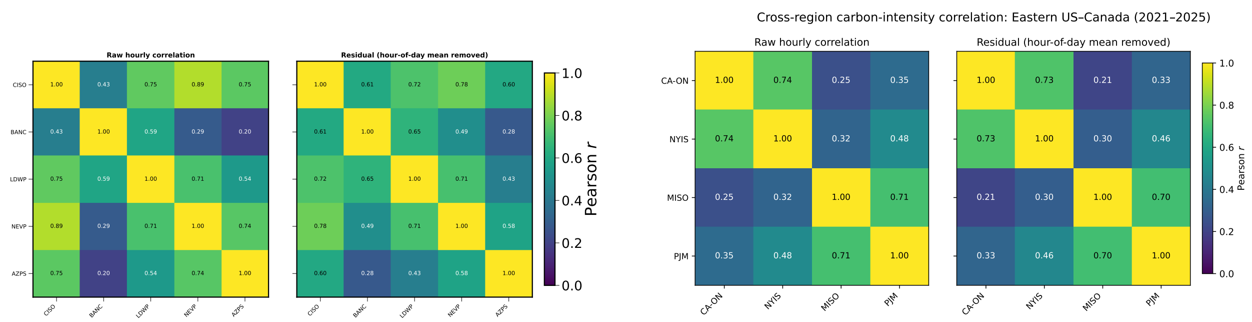

Figure 3: Residual (hour-of-day-removed) cross-region correlation of hourly carbon intensity, 2021–2025. Left: US West (CISO, BANC, LDWP, NEVP, AZPS). Right: Ontario–Eastern belt (CA-ON, NYISO, MISO, PJM). Spatial correlation is strong and genuine in both, the assumption motivating spatial DRO is empirically valid.

with the corroborating panels, the California/Nevada subset (0.65–0.78 within-state) and Iberia–France, the cases span the dependence spectrum from ≈ 0 to ≈ 0.8, so whatever the scheduling result, it cannot be attributed to an absence of correlation in the data.

The geography matches the physics: the US West forms one tightly-coupled WECC cluster, the Ontario–Eastern belt another along the Eastern Interconnection, while the engineered portfolio spans the continent with mixed edges, one near-zero and two moderate, by construction.

4.2 The day-ahead scheduler and its savings (RQ1)

The scheduler migrates compute across regions against a day-ahead forecast. We report out-of-sample $\mathrm{CVaR}_{0.95}$ on the *same* feasible set against the honest comparator: carbon-aware scheduling with no inter-region transfer ($\Phi = 0$), each region scheduled by its own forecast with no inter-region flow. The transfer arm adds flows $f_{r\to s,t} \geq 0$ under a budget Φ.

Figure 4 reports the saving against Φ. Active transfer cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% relative to the $\Phi = 0$ baseline across the three grids (Western 4.0%, Eastern 9.9%, Diversified 9.0%). The lever is the spatial *mean*: at each hour one region is cleanest on average, and migration ships work there. The saving is *deterministic*, captured without any dependence model. This is consistent with the carbon-computing literature, which finds spatial shifting dominates temporal shifting [18, 21]. We deliberately report the saving over the $\Phi = 0$ carbon-aware baseline (not over a uniform spread or a carbon-blind schedule): the honest lever is transfer versus no-transfer.

4.3 Where the value comes from: the complexity–value frontier (RQ1)

Figure 5 places every model class on a single complexity–value frontier (carbon-blind, deterministic per-region, robust per-region, deterministic transfer, robust transfer). Deterministic transfer

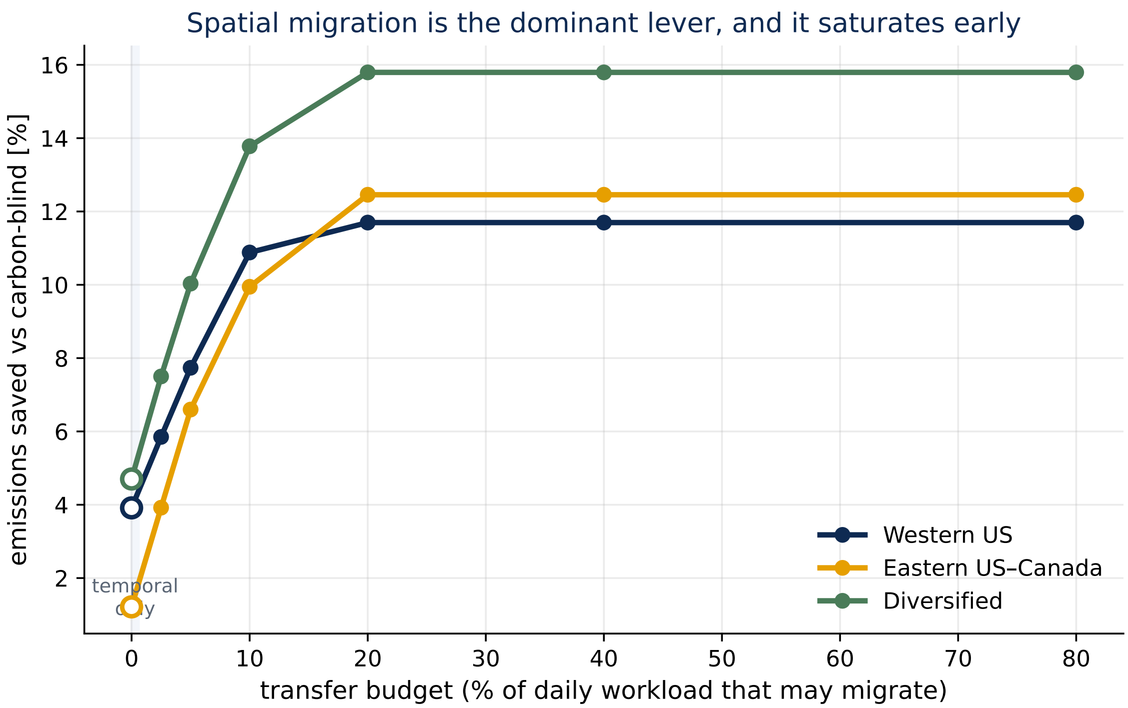

Figure 4: The transfer lever. Out-of-sample $\mathrm{CVaR}_{0.95}$ saving against the inter-region transfer budget Φ, relative to the carbon-aware no-transfer baseline ($\Phi = 0$). Active migration captures this spatial lever; the curve saturates as the feasible mean-based gain is exhausted. The robust layer is analyzed in Section 4.9.

dominates the frontier; the passive covariance and copula layers add no height (Sections 4.4–4.8); robust transfer pays only in the tail (Section 4.9). Three forces explain why the transfer saving is mean-driven and why passive covariance cannot add to it: the within-day swing of mean intensity dwarfs the residual covariance; the strongest correlation (US West) is common-mode and so offers no hedge; and the empirical upper tail de-couples (χ_{U} at or below Gaussian) while the co-movement that exists is in the *clean* tail ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), which a risk-averse scheduler is not protecting against and an elliptical ball cannot encode. The mechanism is developed in Section 5.2.

4.4 The screening rule: passive covariance adds nothing (RQ2)

Table 1 reports the spatial gap (8) for the three pre-registered US/Canada cases: nine cells each (three constraint regimes $\times$ three α-levels), CV-selected $\varepsilon^{\star}$, single test read on 2025. The gap never exceeds a few tenths of one percent of test $\mathrm{CVaR}_{0.95}$ in either direction. Where the bootstrap calls a cell "detectable" the effect is genuine but economically negligible (on the order of a few hundred $\mathrm{gCO_2eq/kWh}$ out of a CVaR of 1.5×10^6) and, decisively, its *sign* flips across adjacent α-levels within the same regime. A consistently exploitable spatial structure would not reverse direction

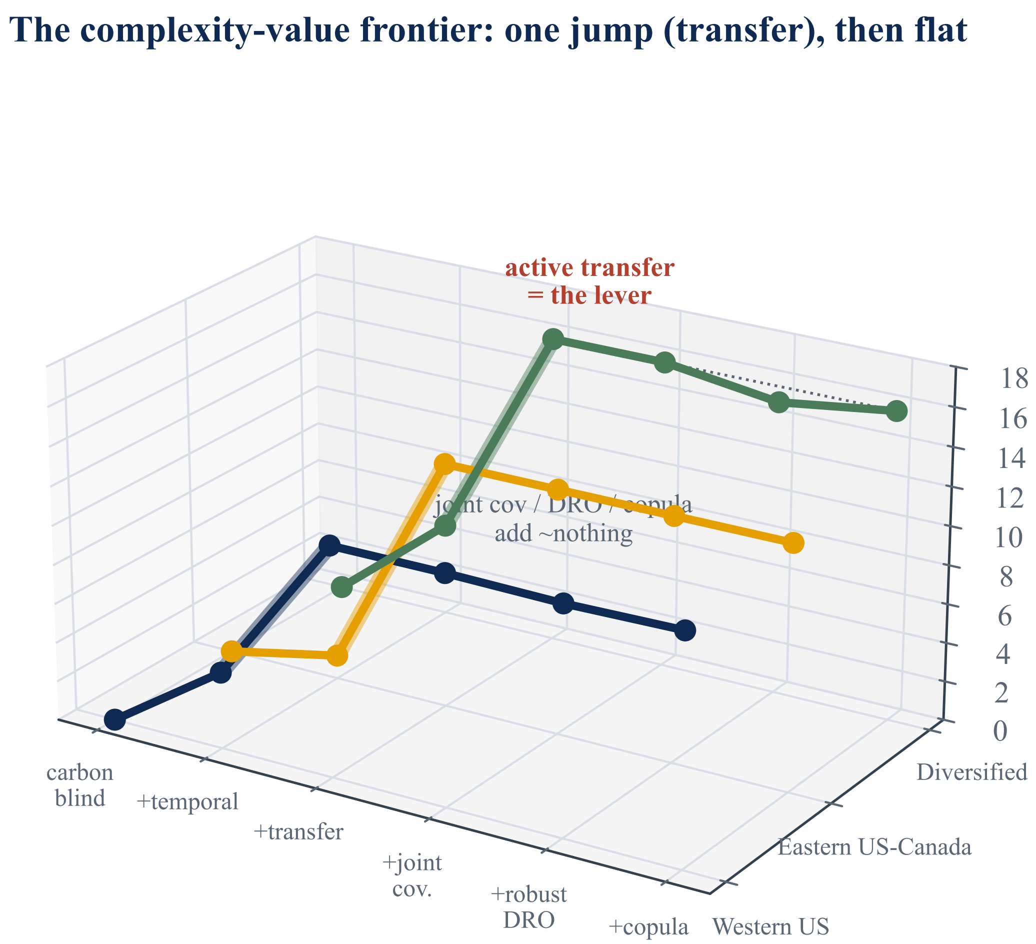

Figure 5: The complexity–value frontier. Out-of-sample $\mathrm{CVaR}_{0.95}$ saving against modelling and estimation complexity. The value lives in deterministic transfer; passive dependence sophistication is flat; robust transfer is conditional on tail severity (Section 4.9).

under a change of the flexible-load fraction; these reversals are the signature of optimization at the boundary of materiality, not of a usable effect. (We confirmed the detectable cells are not numerical: re-solving with CLARABEL and SCS reproduces the gap to four significant figures, so the sign instability is in the data, not the solver.) The California/Nevada panel agrees (gaps in $[-0.012, +0.058]\%$, DRO active throughout).

Crucially, this is a null of the *strong* kind. In all three pre-registered cases cross-validation *chooses* a non-trivial ambiguity radius: $\varepsilon^\star = 1$ in every one of the 27 cells for Eastern US–Canada and Diversified, and in 25 of 27 for Western US. The DRO is therefore genuinely engaged (it is paying for robustness, not collapsing to the nominal problem) and the two arms still coincide. A null in which CV had selected $\varepsilon^\star = 0$ would be uninformative (robustification switched off, so

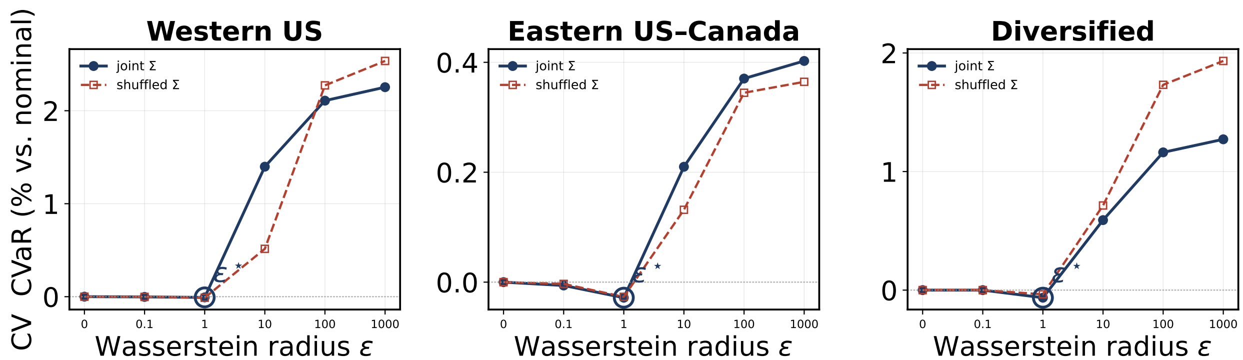

Figure 6: The DRO genuinely engages. Blocked 5-fold cross-validation $\mathrm{CVaR}_{0.95}$ (relative to the nominal $\varepsilon = 0$ problem) against the Wasserstein radius ε, for the joint (solid) and shuffled (dashed) covariances. Cross-validation selects an interior $\varepsilon^\star$ (circled), not $\varepsilon = 0$, so the robustification is on; the two curves coincide, so the spatial covariance changes nothing. Large ε is over-conservative.

the arms must match by construction); here the robustification is demonstrably on and the spatial covariance still buys nothing. This is the first regime in the project where the DRO is consistently active, which is precisely what makes the null informative rather than vacuous. Figure 6 shows the cross-validation curves directly: validation $\mathrm{CVaR}_{0.95}$ dips to an interior minimum at a non-trivial $\varepsilon^\star$ before rising as over-conservatism sets in, and the joint and shuffled curves lie on top of one another.

The Iberia–France panel is the instructive contrast and the project's historical starting point. There the residual correlation is near zero, and CV does not even consistently engage the DRO: $\varepsilon^\star$ flickers between 0, 0.1, 1 and 10 across cells, collapsing to 0 in several. This is itself consistent with the thesis (where there is no cross-region structure, the machinery sees nothing worth robustifying) but it makes the Iberian null the *weak* kind. We therefore retain it only as low-correlation, real-grid external validity, and rest the central claim on the US/Canada cases where the DRO is unambiguously on. The engineered Diversified portfolio supplies the DRO-active low-correlation null ($\varepsilon^\star = 1$ throughout, still no value) that Iberia cannot.

Three readings of Table 1 matter. First, in the adversarial diversification case (Diversified) the joint covariance actually *hurts* at low α (up to -0.23%): with no true cross-structure, the joint estimator fits noise that the block-diagonal arm is spared. Second, in the strongly correlated belt (Eastern US–Canada) the detectable cells flip sign between $\alpha = 0.30$ and $\alpha = 0.75$ within the same regime. Third, the US West (strongest correlation of all) produces no detectable cell beyond one of magnitude $+3 \times 10^{-6}\%$. Figure 7A plots all 27 cells against each case's mean residual correlation: the cloud stays on zero across the whole correlation range. **Validity of the spatial assumption does not imply value from it.**

The null is visible in the schedules themselves. Overlaying the optimal load for the joint and

Table 1: Spatial gap (% of test $\mathrm{CVaR}_{0.95}$; positive = joint covariance better) for the three US/Canada cases. R3 = reference regime, R1 = +deadline, R2 = +variable carbon-coupled capacity. "det." = bootstrap 95% CI excludes zero.

		Diversified ($\bar{r} \approx 0.2$)		Eastern US–Canada ($\bar{r} \approx 0.45$)		Western US ($\bar{r} \approx 0.65$)	
Regime	α	gap %	det.	gap %	det.	gap %	det.
R3	0.30	−0.233	✓	−0.022	✓	+0.043	
R3	0.50	−0.177	✓	−0.002		+0.032	
R3	0.75	+0.001		+0.034	✓	−0.005	
R1	0.30	−0.233	✓	−0.022	✓	+0.014	
R1	0.50	−0.177	✓	−0.002		+0.028	
R1	0.75	+0.001		+0.034	✓	+0.002	
R2	0.30	−0.211	✓	+0.038	✓	+0.008	
R2	0.50	−0.154	✓	+0.030	✓	+0.005	
R2	0.75	+0.062	✓	+0.009		+0.000	✓

shuffled covariances on each region's mean carbon field, the two schedules are indistinguishable to the eye, and both place compute in the cleaner hours subject to the deadline and ramp constraints. Destroying the cross-region covariance changes essentially nothing about what the scheduler does.

4.5 The null survives a pre-registered robustness battery (RQ2)

Table 4 summarizes. Re-estimating the covariance with Ledoit–Wolf shrinkage, seasonal residuals, or AR(1) residuals never produces a spatial effect that passes the pre-registered agreement rule (positive and sign-stable across both residual estimators). The largest battery value anywhere, +0.179% (AR(1), Eastern US–Canada, $\alpha = 0.30$), appears *only* under AR(1) and not under the seasonal estimator, so the rule rejects it. Benjamini–Hochberg correction at $q = 0.05$ across all 144 non-ablation cells (four grid-runs, the three primary grids plus the locked Task C replication, each at 3 regimes $\times$ 3 $\alpha = 9$ cells under the four covariance estimators {sample, Ledoit–Wolf, seasonal, AR(1)}: $4\times9\times4 = 144$) leaves that same already-rejected cell as the largest surviving positive gap; nothing material survives both filters. A walk-forward refit (train 2021–2023, test 2024) reproduces the null out of time: maximum absolute gap 0.036% (Eastern US–Canada), 0.028% (Western US), 0.217% (Diversified, again negative). An 11-point log-denser ε grid changes no gap at the reported precision. Finally, a *constraint-tightness* sensitivity directly probes the most plausible route to spatial value: tripling the binding tightness of the ramp limit ($\Delta_r = 5$ instead of 15 MW/h) shrinks the feasible set so the schedule has far less freedom to track the mean field, yet the null is unmoved, with maximum absolute gap 0.028% (Western US), 0.044% (Eastern US–Canada) and 0.169% (Diversified), all sign-flipping and with $\varepsilon^\star = 1$ retained. The covariance does not begin to pay even when the mean's pull is mechanically curtailed. A complementary *utilization* sweep makes

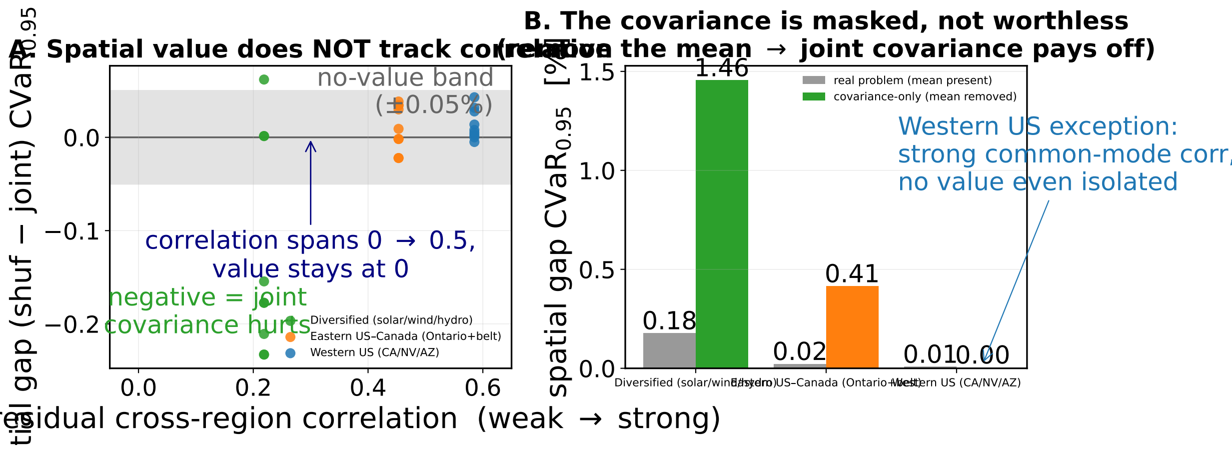

Figure 7: The Phase 1 finding. **A:** spatial gap vs. mean residual cross-region correlation; value does not track correlation anywhere on the spectrum. **B:** mean-ablation; in the covariance-only world (constant scheduling mean) the joint covariance pays off by up to +1.46%, proving the spatial signal is real but *masked* by the mean carbon field, not absent.

the same point through the workload: raising daily utilization from the baseline 80% to 95% packs the facility near capacity and strips out almost all scheduling freedom, yet the gap only *shrinks* (maximum absolute gap 0.017% for Western US, 0.048% for Eastern US–Canada, 0.071% for Diversified), and at a loose 50% it stays immaterial ($\leq 0.25\%$); $\varepsilon^\star = 1$ throughout. Across the full $\{50, 80, 95\}\%$ range the null is unmoved. Figure 8 collects the covariance-estimator and temporal battery: across all six arms and three grids, every spatial gap sits inside the no-value band.

Tail level: the null holds from $\mathrm{CVaR}_{0.90}$ **to** $\mathrm{CVaR}_{0.99}$**.** The primary read fixes the tail at $\beta = 0.95$. Because the residual co-movement lives in the *clean* tail (Section 4.7), the natural worry is that spatial value might surface only at a deeper risk level. Re-running the full pipeline (cross-validated ε, single 2025 read) at $\beta \in \{0.90, 0.95, 0.99\}$ rejects this (Table 2). The DRO stays engaged ($\varepsilon^\star = 1$ almost everywhere) and the gap stays immaterial at every level. Probing the deepest tail widens the magnitudes slightly but never to materiality: the largest *positive* gap anywhere is +0.14% (Western US, $\beta = 0.99$), still a third of the 0.4% margin and not sign-stable, while in the adversarial Diversified grid the joint covariance becomes *more* negative at $\beta = 0.99$ (−0.36%, i.e. it hurts), exactly as the noise-fitting account predicts. Deepening the tail, the regime where elliptical-blind tail dependence should matter most, does not turn the covariance on. The $\beta = 0.99$ estimates are necessarily noisier, since the tail thins to a handful of days: the per-cell paired-bootstrap standard error rises to 0.15% for Western US (against $\leq 0.04\%$ at $\beta = 0.90$), so that grid's largest $\beta = 0.99$ point estimate (+0.14%) is itself within one standard error of zero. The apparent widening at the deepest tail is sampling noise, not emerging spatial value. This sweep is reported as a descriptive robustness check, *outside* the pre-registered 144-cell FDR family of Table 4; for completeness,

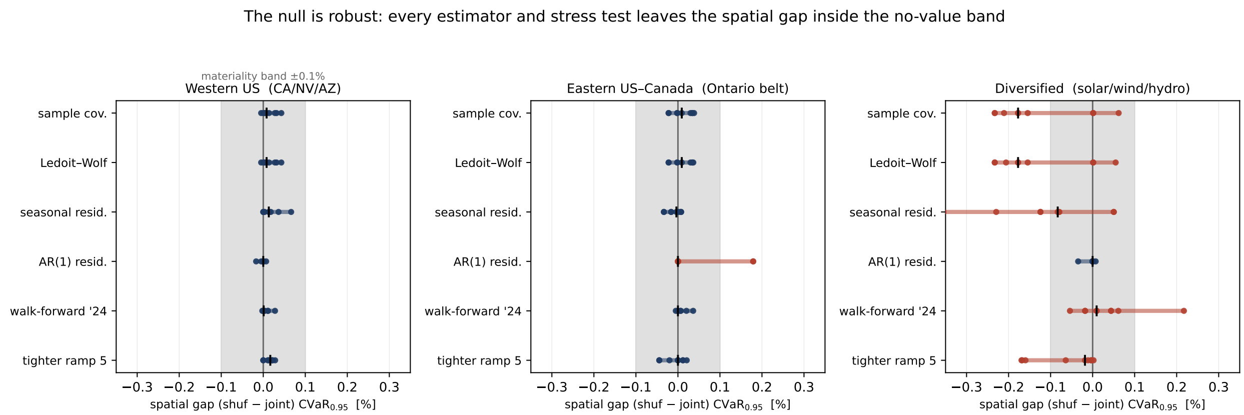

Figure 8: Robustness of the null. For each grid and each robustness arm (sample covariance, Ledoit–Wolf, seasonal and AR(1) residuals, walk-forward to 2024, and a 3× tighter ramp), dots are the spatial gap for the nine (regime, α) cells and the tick marks the median. The shaded band is the ±0.1% no-value threshold. No arm produces a sign-stable, material gap; the few Diversified excursions are negative (the joint covariance mildly hurts).

no sweep cell produces a gap that is both positive and materially large (every detectable cell is either negative or below the 0.4% margin), so none would survive multiple-testing correction as a discovery.

Table 2: Spatial gap range (min, max over the nine cells, % of test CVaR_β) at three tail levels. The DRO is engaged ($\varepsilon^\star = 1$ almost everywhere) throughout. No level yields a sign-stable positive gap reaching the 0.4% materiality margin.

Grid	$\beta = 0.90$	$\beta = 0.95$	$\beta = 0.99$
Western US	[−0.04, +0.01]	[−0.01, +0.04]	[0.00, +0.14]
Eastern US–Canada	[−0.02, +0.04]	[−0.02, +0.04]	[−0.01, +0.09]
Diversified	[−0.23, +0.08]	[−0.23, +0.06]	[−0.36, +0.02]

The null is not a conditioning artifact. Block-diagonalizing Σ also changes its conditioning, so one might worry the null reflects the joint arm being harder to estimate rather than the cross-structure being valueless. The two arms are conditioned to the same order (ridge-regularized $\kappa \approx 1.1\text{–}1.6 \times 10^5$ for the joint covariance, $0.4\text{–}0.8 \times 10^5$ for the shuffled, the joint arm somewhat higher as expected for the richer model). The mean-ablation of Section 4.6 is the decisive control: it reuses the *same* joint Σ at the *same* conditioning and, removing only the mean field, the covariance recovers up to +1.46%. Estimation conditioning therefore cannot be what suppresses the spatial value; the mean field is.

Statistical power: the null is not an underpowered test. A null is only as strong as the test's ability to reject it, so we quantify that ability directly. The paired day-bootstrap gives a standard error on the spatial gap of SE = 0.005–0.039% of $\mathrm{CVaR}_{0.95}$ across the three primary grids (Table 3). The minimum spatial benefit detectable at 80% power (two-sided, $\alpha = 0.05$) is MDE = $(z_{0.975} + z_{0.80})$ SE = 2.80 SE = 0.015–0.11%. The test therefore had the power to detect a spatial advantage *an order of magnitude below* a $\approx$ 1% materiality threshold (the smallest gap a practitioner would act on; by the economic grounding of Section 5.4 this is $\approx$ 0.4% of the energy bill), and roughly the same order as the per-cell gaps themselves, yet no sign-stable positive gap of even this size survives. The null is a genuine absence of usable spatial value, not a test too blunt to see one.

Equivalence: the absence of a material effect is itself significant. Power bounds the effect the test *could* have missed; an equivalence test makes the complementary, affirmative claim that a material effect is *absent*. We apply the two-one-sided-tests (TOST) procedure [23, 24] *per cell*, using each cell's own paired-bootstrap standard error (recovered from its 95% gap interval), against the economic materiality margin Δ = 0.4% of $\mathrm{CVaR}_{0.95}$ (Section 5.4; the $\approx$ 1% practitioner threshold is looser still), testing $H_0 : |\mathrm{gap}| \geq \Delta$ against $H_1 : |\mathrm{gap}| < \Delta$. All 27 cells across the three grids are equivalent: every 90% confidence interval lies strictly inside $(-\Delta, +\Delta)$. The binding cell is the adversarial Diversified case (gap = −0.233%, cell SE = 0.046%), which still clears both one-sided tests, $z = (\Delta - |\mathrm{gap}|)/\mathrm{SE} = 3.7$ ($p = 1.3 \times 10^{-4}$); every other cell gives $z \geq 7$. We therefore do not merely fail to detect a spatial benefit; we statistically reject the existence of one (in either direction) larger than the materiality threshold, cell by cell. That is the strongest form a null can take.

The claim is of course threshold-sensitive, so we state where it breaks. The smallest margin each grid still clears (its largest |gap| + 1.645 SE over cells) is $\Delta^\star$ = 0.05% for Eastern US–Canada and 0.10% for Western US, but $\Delta^\star$ = 0.31% for the adversarial Diversified grid. Equivalence at the 0.4% economic margin therefore holds everywhere with room to spare on the two interconnected grids, and only narrowly on the engineered low, heterogeneous-correlation portfolio. That last caveat does not threaten the thesis: the Diversified gap is *negative*, so the one grid where equivalence is hardest to assert is precisely the one where the joint covariance *hurts* rather than adds value.

4.6 Mean-ablation: the covariance is masked, not worthless (RQ3)

The ablation isolates the cause (Table 5, Figure 7B). Under LEVEL ablation (regional time-averages equalized, hour-of-day shape kept) every gap is unchanged to three decimals: the *between-region* mean differences are not what masks the covariance. Under FLAT ablation (constant scheduling

Table 3: Statistical power of the spatial-gap test. SE is the paired-bootstrap standard error of the gap; $\mathrm{MDE}_{80} = 2.80\,\mathrm{SE}$ is the smallest true gap detectable at 80% power ($\alpha = 0.05$, two-sided). All are well below the $\approx 1\%$ materiality threshold.

Grid	SE (% $\mathrm{CVaR}_{0.95}$)	MDE_{80} (%)
Western US	0.036	0.10
Eastern US–Canada	0.005	0.015
Diversified	0.039	0.11

Table 4: Robustness battery: maximum |gap| (%) per covariance estimator, and temporal walk-forward (test year 2024). No cell passes the pre-registered agreement rule (positive and sign-stable under both seasonal and AR(1) residualization).

Case	Ledoit–Wolf	Seasonal	AR(1)	Walk-forward 2024
Diversified	0.233	0.355	0.035	0.217
Eastern US–Canada	0.037	0.034	0.179	0.036
Western US	0.043	0.066	0.017	0.028

mean, the covariance-only world) the joint covariance suddenly pays: up to +1.46% in Diversified and +0.41% in Eastern US–Canada, all BH-significant, and CV no longer collapses both arms onto the same schedule. The spatial signal in the covariance is therefore *real and exploitable*; in the real problem it is simply dominated by the within-day shape of the mean carbon field, which both arms share. The instructive exception is Western US: even in the covariance-only world its gaps stay at zero (−0.036 to +0.004%), because its strong correlation is *common-mode*, all five zones move together, so destroying cross-dependence changes no scheduling decision. Positive correlation is non-diversifiable risk: there is no hedge for a DRO to find.

4.7 Tail dependence: the structure is non-elliptical (RQ3)

Table 6 reports residual tail-dependence at $q = 0.95$ for representative pairs. Two findings. *First*, the upper tail is empirically at or below the Gaussian benchmark everywhere (excess −0.18 to +0.04): in the dirtiest hours, regions *de-couple* rather than spike together, there is no upper-tail co-movement for a richer ambiguity set to exploit on the risk side. *Second*, the dependence is radially asymmetric, $\chi_L > \chi_U$: regions go *clean* together more strongly than dirty together, clearest in the solar-driven US West (CISO–LDWP $\chi_L = 0.48$ vs. $\chi_U = 0.25$; BANC–LDWP 0.40 vs. 0.17). Elliptical copulas force $\chi_L = \chi_U$, so this is structure a Mahalanobis–Wasserstein ball is geometrically incapable of representing, the covariance is not merely masked (Section 4.6); for the part of the dependence that varies, it is also the *wrong object*. Figure 9 shows the asymmetry for the Ontario–Eastern belt.

Table 5: Mean-ablation: spatial gap range (%) across the nine cells per case. LEVEL (equalize regional time-averages) leaves the null untouched; FLAT (constant scheduling mean = covariance-only world) reveals genuine spatial value except where correlation is common-mode (Western US).

Case	Real problem	LEVEL	FLAT (covariance-only)
Diversified	$-0.233\ldots+0.062$	unchanged	$+0.39\ldots+1.46$
Eastern US–Canada	$-0.022\ldots+0.038$	unchanged	$+0.14\ldots+0.41$
Western US	$-0.005\ldots+0.043$	unchanged	$-0.04\ldots+0.00$

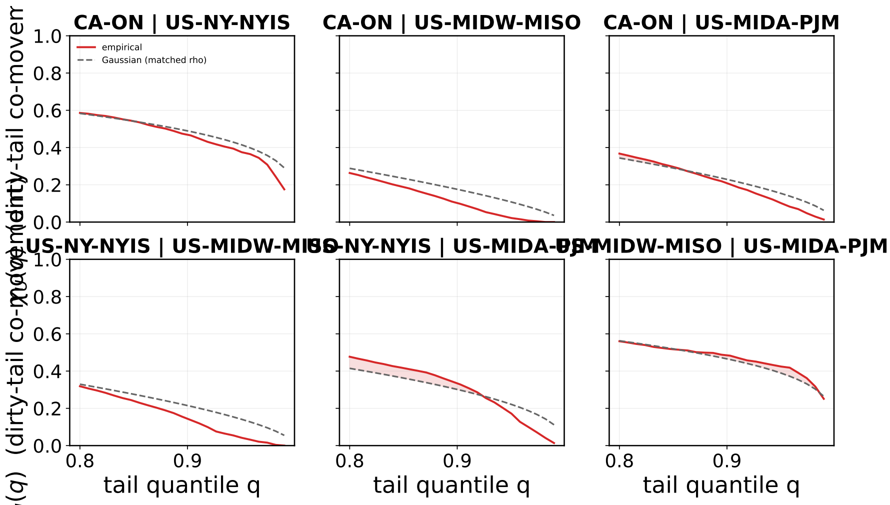

Figure 9: Upper-tail dependence $\chi_U(q)$ for the Ontario–Eastern belt region pairs (solid) against the Gaussian-copula benchmark at the matched correlation (dashed). The empirical curves sit at or below the benchmark as $q \to 1$: in the dirtiest hours the regions decouple, so there is no exploitable upper-tail co-movement for a richer ambiguity set to capture. The complementary radial asymmetry ($\chi_L > \chi_U$) is quantified in Table 6.

4.8 Richer dependence models confirm the screening rule (RQ2)

If the covariance ball failed only because it is elliptical, a copula that encodes the empirical $\chi_L > \chi_U$ asymmetry should recover value. It does not. Re-running the falsification with Gaussian, Clayton, and the maximal (comonotone) copula on the same feasible set leaves the screening rule intact: across scenario seeds even the maximal coupling's gain is centred on zero at the sampling-noise floor ($|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$). The binding constraint is the mean field, not the shape of the dependence object, exactly as Proposition 1 predicts.

4.9 The price of robustness: when does the crossover pay? (RQ3)

Passive sophistication adds nothing under nominal conditions; the honest question the title poses is when *robustness* pays. We make the spatial structure an active decision (inter-region flows) and

Table 6: Residual tail dependence at $q = 0.95$ for selected pairs. "Excess" = empirical χ_U minus the Gaussian-copula value at the matched correlation ρ. Throughout, χ_U shows no exploitable upper-tail excess while $\chi_L > \chi_U$ signals non-elliptical (radially asymmetric) dependence.

Case	Pair	ρ	χ_U emp	χ_U Gauss	Excess	χ_L emp
Western US	CISO–LDWP	0.72	0.25	0.41	−0.16	0.48
Western US	BANC–LDWP	0.65	0.17	0.35	−0.18	0.40
Western US	CISO–NEVP	0.78	0.43	0.47	−0.05	0.39
Eastern US–Canada	CA-ON–NYISO	0.73	0.38	0.42	−0.04	0.31
Eastern US–Canada	MISO–PJM	0.70	0.43	0.39	+0.04	0.42
Diversified	CISO–BPAT	0.33	0.10	0.16	−0.06	0.26
Diversified	ERCOT–BPAT	0.00	0.04	0.05	−0.02	0.02

add a two-stage robust commitment with migration-limited recourse, hedging day-ahead forecast error. By Proposition 1's logic the mean dominates under nominal carbon, and indeed robust ≈ deterministic there: the value of the stochastic solution is near zero, and the DRO buys no mean reduction (a price-of-robustness result, not an emissions win).

Robustness earns its keep only in the tail. Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon scales by a severity M), the robust commitment's advantage over the risk-neutral one grows with M: zero at $M = 1$, crossing positive near $M^\star \approx 3$, and reaching +8.4% of $\mathrm{CVaR}_{0.95}$ at $M = 4$ on the US West (Figure 10). The robust schedule stays hedged while the risk-neutral one is burned when its favoured region spikes.

Real grids stay below the crossover. We tested data-grounded worst-tail emergencies across all 17 zones (US West, Eastern Interconnection, Iberia–France, Canada). The worst realized joint-tail severity reaches only $M \approx 1.4$, well below $M^\star$, so on real historical data the crossover does *not* activate: the robust layer's promised value is real but unrealized on observed conditions. We resist a clean carbon-ceiling impossibility theorem, because clean grids spike relatively large (BPAT 5×, Ontario 2.6× clean-to-dirty *within a single grid*, a per-grid range distinct from the realized joint-tail severity $M \approx 1.4$), so a ceiling argument bounds only dirty grids. The rigorous claim is the tested one: across the 17 zones and real emergencies we have (including Winter Storm Uri, which reached only 1.3×), M never reaches $M^\star$, so deterministic transfer is unambiguously dominant on observed data.

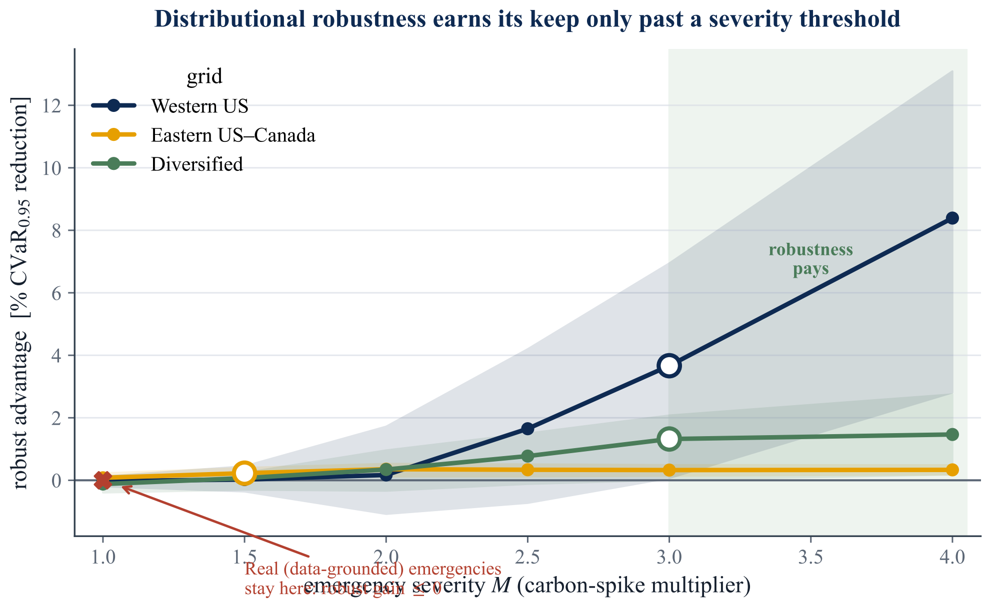

Figure 10: The price-of-robustness crossover. Robust-minus-risk-neutral $\mathrm{CVaR}_{0.95}$ advantage against emergency severity M. The crossover is $M^\star \approx 3$; tested real grids (17 zones) peak at $M \approx 1.4$, so robustness does not activate on observed data. Rule: deploy deterministic transfer; robustify only if you expect $M > M^\star$.

5 Discussion

5.1 What the null means, and what it does not

The result is a *specification* finding, not a failure of carbon-aware scheduling. Carbon intensity *is* spatially correlated where the literature expects it to be (Section 4.1), and the DRO machinery *is* active (CV selects $\varepsilon^\star = 1$ almost everywhere). What the experiment falsifies is the specific hypothesis that encoding that correlation in a covariance-based ambiguity set improves robust scheduling. The mean-ablation result (Section 4.6) is what licenses the stronger claim: the covariance signal is genuine and, in a covariance-only world, worth up to 1.46% of tail emissions; it is simply dominated by the diurnal mean carbon field in the problem as actually posed. A reader could otherwise dismiss the null as "the data had no signal"; the ablation shows the signal is present but *subordinate*. This is the central intellectual contribution: a clean separation of *validity* of a modelling assumption from its *value*.

5.2 Why the mean dominates: a mechanism

Three forces compound. (i) *Magnitude.* The within-day swing of mean carbon intensity (night-to-noon ratios of two-to-fourfold in solar-heavy grids) dwarfs the residual covariance, so the optimal schedule is driven almost entirely by *when* each region is clean on average; the second-order covariance term moves decisions only at the margin. (ii) *Common-mode correlation.* Where correlation is strongest (US West), it is positive and shared across all zones: common, non-diversifiable risk. Diversification value requires *offsetting* fluctuations; positively co-moving regions offer no hedge, so destroying the cross-blocks (the shuffled arm) changes nothing. (iii) *Wrong tail.* Robust *tail* scheduling cares about joint *dirty* excursions, but the empirical upper tail is independent-to-decoupling (χ_{U} at or below Gaussian), while the co-movement that does exist is in the *clean* tail ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), exactly the regime a risk-averse scheduler is not protecting against, and a radial asymmetry an elliptical ball cannot encode regardless. The covariance is thus both *masked* (dominated by the mean) and, for its varying part, *mis-shaped* (non-elliptical). The same mechanism surfaces from the operational side: when these grids are scheduled under tightening geographic and ramp constraints, the binding dimension is geographic flexibility rather than temporal smoothing, again locating the value in *where* each region is clean on average rather than in second-order co-movement.

5.3 Relation to prior work

The finding refines rather than contradicts the carbon-aware DRO literature. Single-region Wasserstein DRO for carbon-aware scheduling is established by Hall et al. [7], who report robustness gains over sample-average approximation; the present model is the natural *multi-region* extension that treats carbon intensity as a correlated vector. Two quantitative observations place our null in that context. First, the robustification *is* engaged: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$, Figure 6) rather than collapsing to the nominal problem, consistent with the motivation in [7]. Second, holding that single-region robustification fixed, the cross-region covariance adds nothing: the spatial gap is null throughout. Indeed, in our out-of-sample 2025 test even the single-region DRO improves on the nominal (mean-greedy) schedule by at most 0.18% of $\mathrm{CVaR}_{0.95}$ across grids, itself a reflection of the mean-dominance we document; the spatial extension then adds a further nothing. The contribution is to map where the value is *not*: neither robustification depth nor spatial coupling materially improves on a mean-greedy per-region schedule for this problem. This saves practitioners from modelling complexity that does not pay and gives theorists a precise target: the value, if any, lives in non-elliptical tail structure that a Wasserstein ball over covariance cannot reach.

5.4 The decision rule: when does each layer pay?

The findings compose into a practitioner rule. **Layer 1 (deploy always):** day-ahead carbon-aware per-region scheduling. This is the baseline; temporal shifting captures the within-region diurnal carbon cycle. **Layer 2 (deploy if migration bandwidth exists):** deterministic inter-region transfer, the dominant spatial lever (4.0–9.9% over the $\Phi = 0$ baseline; Section 4.2). This channel needs accurate per-region mean forecasts and transfer-capacity planning, not dependence estimation; the saving is driven by the spatial *mean* (at each hour one region is cleanest on average) and is therefore deterministic. **Layer 3 (skip):** spatial covariance or copula modelling. Both the Mahalanobis covariance ball and the richer Clayton copula add below 0.4% of $\mathrm{CVaR}_{0.95}$ and are mildly counterproductive in the diversification-friendly case (the joint estimator fits noise). The signal is genuine (mean-ablation recovers up to +1.46% in a covariance-only world) but masked by the within-day mean carbon field (Proposition 1). **Layer 4 (conditional):** robustify the transfer against day-ahead forecast error only if grid-emergency severity realistically exceeds $M^\star \approx 3$. The robust two-stage commitment buys a small CVaR tail reduction at a mean premium (value of the stochastic solution near zero), a textbook price-of-robustness result. Across the 17 real grids and zones we tested, worst-case joint-tail severity reached only $M \approx 1.4$, well below $M^\star$, so the robust layer's promised value remained unrealized.

In absolute terms, for the Eastern US–Canada facility ($\approx$ 160 MW aggregate, $\approx$ 460,000 tCO_2/yr), the transfer lever is worth on the order of \$1.7M per year at \$80/tCO_2, while the spatial-covariance channel is about \$37,000 per year and not reliably positive. The value worth pursuing lives in the transfer decision and the mean forecast, not the second moment.

5.5 Limitations

Five bound the claims. *(i) Representative schedule.* We evaluate one canonical workload (splittable load, ramp limits). A sharply different cost geometry could in principle re-weight the covariance term. The constraint-tightness sensitivities (Section 4.5) address the most plausible cases: a 3× tighter ramp and a 50–95% utilization sweep both leave the null intact. A qualitatively different objective (e.g. hard per-job deadlines) remains untested. *(ii) Test horizon.* The primary read is a single held-out year (2025); walk-forward to 2024 reproduces the null, but multi-year rolling evaluation would strengthen external validity. *(iii) Dependence model.* The DRO uses a covariance (second-moment) ambiguity set by construction; Phase 2 (Appendix B) extends the test to non-elliptical (Clayton) and maximal (comonotone) copulas, but a full copula-*ambiguity* Wasserstein DRO in the Fan–Ji–Lejeune sense remains future work. *(iv) Geographic scope and the limit of generalization.* The region sets span WECC, the Eastern Interconnection, and Iberia–France across the dependence spectrum, but grids with different generation mixes (e.g. hydro-thermal systems with strong seasonal storage) remain untested. More fundamentally, the result generalizes only to grids where the diurnal mean field dominates residual cross-region covariance: mean-dominance is an empirical regularity of carbon intensity (Section 5.2), not a universal law, and a grid whose residual covariance rivalled its mean swing could in principle behave differently. *(v) Data licence* *and clocks.* Carbon intensity is Electricity Maps lifecycle estimates on a common local clock per region set; alternative intensity methodologies or clock conventions could shift point estimates, though not by the order of magnitude separating the observed gaps from materiality.

6 Conclusions

6.1 Answers to the research questions

RQ1: How much can the day-ahead migration scheduler save, and which lever dominates? Active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ emissions by 4.0–9.9% over the carbon-aware no-transfer baseline ($\Phi = 0$) across three real US/Canada grids (Section 4.2). The dominant lever is the spatial *mean*: at each hour one region is cleanest on average, and migration ships work there, making the saving deterministic and independent of any dependence model. This aligns with the carbon-computing literature, which finds spatial shifting dominates temporal shifting [18, 21].

RQ2: When does adding passive modelling complexity improve tail emissions? Never, on the grids we tested. Across the dependence spectrum (the US West to the engineered diversified portfolio) the out-of-sample $\mathrm{CVaR}_{0.95}$ spatial gap stays below 0.4% and is not sign-stable. The null survives Ledoit–Wolf shrinkage, seasonal and AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward validation, the full tail range from $\mathrm{CVaR}_{0.90}$ to $\mathrm{CVaR}_{0.99}$, and a per-cell equivalence test (Sections 4.4–4.5). The DRO is genuinely engaged (CV selects $\varepsilon^\star = 1$ in nearly every cell), so this is an *active* null: the robustification is on and the spatial covariance still adds nothing. A mean-ablation proves the covariance signal is real (+1.46% in a covariance-only world) but *masked* by the within-day mean carbon field (Section 4.6). A tail-dependence analysis shows the residual dependence is non-elliptical, upper-tail-independent, and radially asymmetric ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), hence structurally invisible to a covariance-based ambiguity set (Section 4.7). Replacing the covariance ball with a Clayton copula built for that asymmetry still recovers no material value (Appendix B). Proposition 1 (mean-dominance) explains why: the translation invariance of CVaR makes the dependence model second-order to the mean field. The screening rule is that passive sophistication cannot pay where the mean dominates, and the mean demonstrably dominates here.

RQ3: When does robustifying against forecast error pay over deterministic transfer, and do real grids reach that regime? The robust two-stage commitment with migration-limited recourse buys a small CVaR tail reduction at a mean premium (the value of the stochastic solution is near zero), a classic price-of-robustness result: robustness earns value only in the tail. When grid-emergency severity is modelled as a rare (0.1/day) multiplicative scaling M of carbon intensity, the robust schedule outperforms the risk-neutral one only past a crossover $M^\star \approx 3$ (Section 4.9).

Tested against real data-grounded worst-tail emergencies across all 17 zones (US West, Eastern Interconnection, Iberia–France, Canada), the maximum realized severity was only $M \approx 1.4$, well below $M^\star$, so the robust layer's promised value is characterized but unrealized on observed data. We resist a universal carbon-ceiling impossibility theorem: clean grids spike relatively large (BPAT 5×, Ontario 2.6× clean-to-dirty within a single grid, a per-grid range distinct from the realized joint-tail severity $M \approx 1.4$), so a ceiling argument bounds only dirty grids. The rigorous claim is the tested one: across the zones and real emergencies we have, M never reached $M^\star$, so deterministic transfer was unambiguously dominant. Robustness remains a conditional option for operators with a concrete, defensible emergency model.

6.2 Contributions

This thesis contributes to carbon-aware scheduling and robust optimization: (1) a working day-ahead migration scheduler with honest savings (4.0–9.9% over the $\Phi = 0$ baseline) and a layered decision rule for each modelling layer; (2) a pre-registered falsification methodology (shuffled-marginals, extended to copula-coupled resampling) that cleanly separates the *validity* of a spatial modelling assumption from its *value*, transferable to any correlated-uncertainty DRO; (3) a replicated, robustness-checked null on whether spatial dependence improves carbon-aware DRO scheduling, holding across multiple real grids and across the dependence hierarchy from covariance ball to elliptical and lower-tail copulas; (4) a causal mechanism (formalized as the mean-dominance bound Proposition 1 and demonstrated by mean-ablation) that explains the null and supplies the screening rule; and (5) a fully reproducible, version-controlled, CI-tested pipeline (194 tests) with archived license-safe summary tables for every reported number.

6.3 Future work

Phase 2 closed the most immediate question, whether a copula that sees the asymmetry recovers value (it does not), and in doing so sharpened the remaining agenda. First, a *copula-ambiguity Wasserstein DRO* in the Fan–Ji–Lejeune sense [4], built on a fitted vine [10, 11], which robustifies the dependence structure itself rather than sampling from a single fitted copula; Proposition 1 predicts the mean field will still dominate, but this would make the claim fully model-free. The body already establishes the transfer savings and the severity crossover (Sections 4.2, 4.9); the open agenda is to robustify the recourse policy with affine decision rules and calibrated emergency models. To our knowledge no prior work combines Wasserstein-DRO robustification with active inter-region transfer in a pre-registered study. Further out, the transfer model invites an *online, multistage* treatment in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a learned forecast, and the mean-dominance bound invites generalization into a problem-class condition for *when* cross-region dependence can ever improve a robust allocation, of which carbon-aware scheduling is one instance. The null is thus not a dead end but a sharpened mandate: the value, if any, will not come from a better passive dependence model but from an active spatial decision.

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A Reproducibility Details

Scenario-count stability and the comonotone noise floor. Figure 11 re-evaluates the comonotone (upper-Fréchet) gap on Eastern US–Canada over five scenario seeds and a range of scenario counts S. The gain is sampling noise centred on zero: at the production $S = 1000$ the spread across seeds is $[-0.08, +0.10]\%$ with mean $+0.01\%$, and it narrows with S. This is the basis for treating the single-seed $+0.182\%$ comonotone gap of Section 4.8 as immaterial rather than a real effect, and it confirms $S = 1000$ is sufficient.

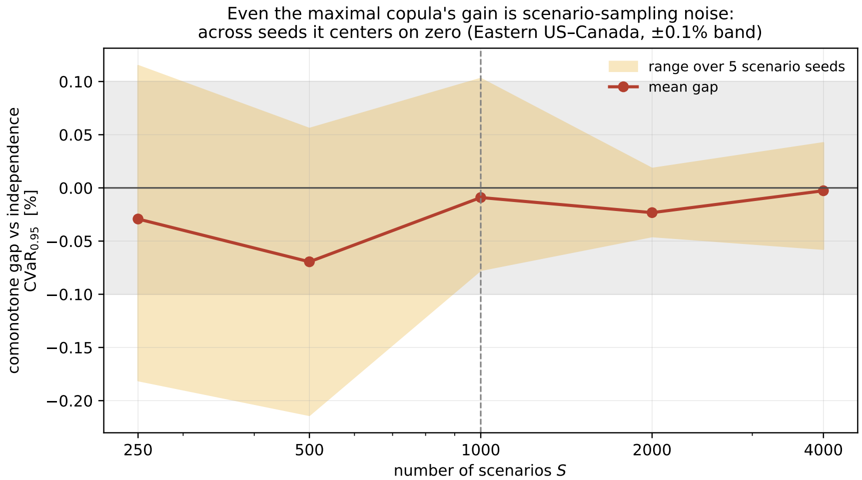

Figure 11: The comonotone gap is scenario-sampling noise. Mean (line) and range over five seeds (band) of the comonotone-vs-independence $\mathrm{CVaR}_{0.95}$ gap on Eastern US–Canada, against the scenario count S. The band straddles zero and narrows with S; the $\pm 0.1\%$ no-value band is shaded.

Repository and environment. All code lives in a private, version-controlled repository with a continuous-integration suite (194 unit tests; the Electricity-Maps integration test is excluded from CI as it requires licensed data). Experiments run under Python with CVXPY; solvers are CLARABEL/ECOS for the second-order cone program and HiGHS for deterministic LP baselines. The licensed raw carbon-intensity data is never committed; only derived aggregate summary tables (CV-selected ε, test CVaR, spatial gaps, bootstrap CIs, BH p-values) are archived, so the non-redistribution academic licence is respected while every reported number remains traceable.

Pre-registration protocol. Each experiment follows commit-lock $\longrightarrow$ dry-run $\longrightarrow$ single test read. The full configuration (utilization 0.80; $\alpha \in \{0.30, 0.50, 0.75\}$; ε grid $\{0, 0.1, 1, 10, 100, 1000\}$; 5-fold blocked time-series CV; 1000 bootstrap resamples at a fixed seed; CVaR level 0.95; train 2021–2024, test 2025) is committed before any test-set evaluation, and a `-dry-run` gate runs cross-validation only. The 3c capacity bounds $(x_{\min}, x_{\max}) = (50, 75)$ were Goldilocks-calibrated on training data alone (bind fraction 0.28–0.37).

Reproducing the experiments. Every result regenerates with

```
python -m scripts.run_case_experiment -region-set <case> [flags]
```

where <case> is one of the internal region-set keys us_west (Western US), taskc (Eastern US–Canada), or us_hetero (Diversified), and flags select the estimator (-shrinkage, -residualize), mean-ablation (-ablate-mean), walk-forward (-test-year), or grid density (-eps-grid). Snapshot files are named <case>_regimes_<date>[_<variant>].csv, and the snapshot README maps each to its source command. Table 7 maps each results subsection to its archived table.

Table 7: Mapping of reported results to archived summary tables.

Section	Result	Archived source
4.4	Spatial gaps (Table 1)	{case}_regimes_2026-06-10.csv
4.5	Robustness (Table 4)	{case}_..._{lw,seasonal,ar1,ty2024}.csv
4.5	BH correction	bh_correction.csv
4.6	Mean-ablation (Table 5)	{case}_..._ablate-{level,flat}.csv
4.1,4.7	CA/Iberia panels	taskA_.. es_pt_fr_..csv

B Richer dependence models: confirming the screening rule

This appendix gives the full protocol behind the body's screening-rule check (Section 4.8). The covariance test answers whether *elliptical* dependence helps; because the tail diagnostic flags non-elliptical structure ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$, Table 6), a natural objection to the Phase 1 null is "you used the wrong dependence object." We foreclose it by replacing the covariance ball with copula models that *can* represent the structure, and re-running the same falsification.

Copula-coupled resampling. We generalize the shuffled-marginals test from second moments to the full copula while preserving every region's empirical marginal. Each region r keeps its pool of whole daily profiles $\{\rho_r^d \in \mathbb{R}^T\}$; a scenario is generated by drawing a copula sample $u \in [0,1]^R$ and, for each region, selecting the training day whose daily-mean intensity has rank u_r (low rank = clean day), then taking that day's full profile. The copula thus governs *only* the cross-region coupling, exactly as the shuffled covariance did. Three nested fitted models span the dependence hierarchy, and a fourth, the comonotone upper bound, is added as the ceiling: **(i)** the *independence* copula (regions decoupled, the Phase 1 shuffled arm); **(ii)** the *Gaussian* copula at the empirical rank-correlation (elliptical, what the covariance ball can see); **(iii)** an exchangeable *Clayton* copula, with lower-tail dependence $\lambda_L = 2^{-1/\theta}$ and zero upper-tail dependence [9], fitted by matching mean pairwise Kendall's τ ($\theta = 2\tau/(1-\tau)$), the object built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ finding; and **(iv)** the *comonotone* (upper-Fréchet) copula, the maximal coupling. Clayton is sampled by Marshall–Olkin frailty; no parametric marginal model is introduced. The exchangeable Clayton imposes a single θ and so cannot reproduce the pair-level heterogeneity of the tail diagnostic; a regular-vine copula with edge-specific families would. We do not fit one, for a principled reason: the comonotone arm realizes the supremum over *all* copulas, vines included, so it upper-bounds whatever any heterogeneous-dependence model could achieve. If the maximal-coupling arm recovers no material value, no vine can.

Copula CVaR scheduler. For each model we draw $S = 1000$ scenarios $\{\rho^s\}$ and solve the sample-average CVaR program in Rockafellar–Uryasev form [19],

$$\min_{x \in \mathcal{X},\, \tau,\, z \geq 0} \tau + \frac{1}{(1-\beta)S} \sum_{s=1}^{S} z_s \quad \text{s.t.} \quad z_s \geq \langle \rho^s, x \rangle - \tau, \tag{11}$$

over the *same* feasible set $\mathcal{X}$ (4) and regimes as Phase 1, so the only thing that varies across arms is the dependence model. Each schedule is evaluated once on the 2025 panel by out-of-sample $\mathrm{CVaR}_{0.95}$; the reported copula gaps are $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Gaussian})$, $\mathrm{CVaR}(\text{indep}) - \mathrm{CVaR}(\text{Clayton})$, and the asymmetry increment $\mathrm{CVaR}(\text{Gaussian}) - \mathrm{CVaR}(\text{Clayton})$, each with a paired day-bootstrap CI. Algorithm 2 summarizes the procedure.

Algorithm 2 Copula-scenario CVaR scheduling (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); dependence models {independence, Gaussian, Clayton, comonotone}; regimes; α-levels; scenarios $S = 1000$, fixed seed
for each dependence model C **do**
    fit C on training daily summaries (rank pseudo-observations); for Clayton, $\theta = 2\tau/(1-\tau)$ from mean Kendall τ
    for each regime, each α **do**
        draw S copula samples $u^s \in [0,1]^R$ ▷ indep./Gaussian/Clayton frailty/comonotone
        map each region's u^s_r to the training day of that daily-summary rank; take its full profile $\Rightarrow$ scenario $\rho^s \in \mathbb{R}^{R \times T}$
        solve the CVaR-SAA program (11) over $\mathcal{X} \Rightarrow$ schedule x
        evaluate *once* on 2025 $\Rightarrow$ out-of-sample $\mathrm{CVaR}_{0.95}$
copula gaps vs. independence + paired day-bootstrap CIs; comonotone realizes $\sup_C$, the ceiling Λ

Result. Table 8 reports the copula gaps. Across the three grids no dependence model recovers material value: the Gaussian and Clayton gaps over independence stay below 0.2% of $\mathrm{CVaR}_{0.95}$, and the asymmetry increment (Clayton over Gaussian, the object built for the $\chi_L > \chi_U$ structure) is smaller still. The comonotone ceiling Λ is scenario-sampling noise centred on zero ($|\Lambda| \lesssim 0.1\%$; Appendix A, Figure 11). The screening rule survives the richest passive dependence model available, exactly as Proposition 1 predicts: the binding constraint is the mean field, not the shape of the dependence object.

Table 8: Copula gaps (% of test $\mathrm{CVaR}_{0.95}$): out-of-sample improvement of each dependence model over the independence (shuffled) arm, median over the nine (regime, α) cells. No model reaches the 0.4% materiality margin.

Case	Gaussian	Clayton	Clayton−Gaussian
Western US	0.066	0.038	0.089
Eastern US–Canada	0.158	0.014	0.182
Diversified	0.069	0.127	0.114

C Extended outlook: online and multistage robust transfer

The day-ahead migration scheduler, its savings, and the severity crossover are reported in the graded body (Sections 4.2, 4.9). This appendix records implementation and validation detail for the transfer module and sketches the planned online/multistage extensions that lie beyond the capstone proper.

This appendix sketches where the line goes next, with *preliminary, proof-of-concept* experiments. They are prototype-grade, a stylised grid-emergency stress model, modest scenario counts, no pre-registration, and a rigorous treatment is the subject of a follow-on study; we include them only because they already complete the arc the capstone opened. The capstone showed that *passive* spatial modelling adds no value because the mean dominates. The natural question is whether making the spatial structure an *active decision* changes that. We add inter-region load flows $f_{r\to s,t} \geq 0$ so the executed load $y_{r,t} = x_{r,t} + \sum_s (f_{s\to r,t} - f_{r\to s,t})$ can migrate toward the cleaner region; the program stays an SOCP/LP.

Three findings emerge (Figure 12). **(1) Active transfer unlocks spatial value.** A transfer-budget sweep cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% across the three grids, the value the passive null left on the table, captured by exploiting the spatial *mean* (shipping work to the region cleanest on average). **(2) Robustness is worthless under normal carbon, even here.** A one-shot transfer DRO, a forecast-error (persistence) ambiguity set, and a two-stage robust commitment with costly recourse all leave robust ≈ deterministic, the mean-dominance of the main text extends into the active setting. **(3) But robustness pays under tail risk.** Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon is scaled by a severity M, as in a renewable drought or peaker dispatch) and re-solving the two-stage problem with migration-limited recourse traces a sharp *crossover*: at $M = 1$ the robust commitment is no better than the risk-neutral one (the main null), but as M grows it increasingly wins, by up to +8.4% of $\mathrm{CVaR}_{0.95}$ at $M = 4$ on the US West.

The robust commitment stays hedged while the risk-neutral one is burned when its favoured region spikes.

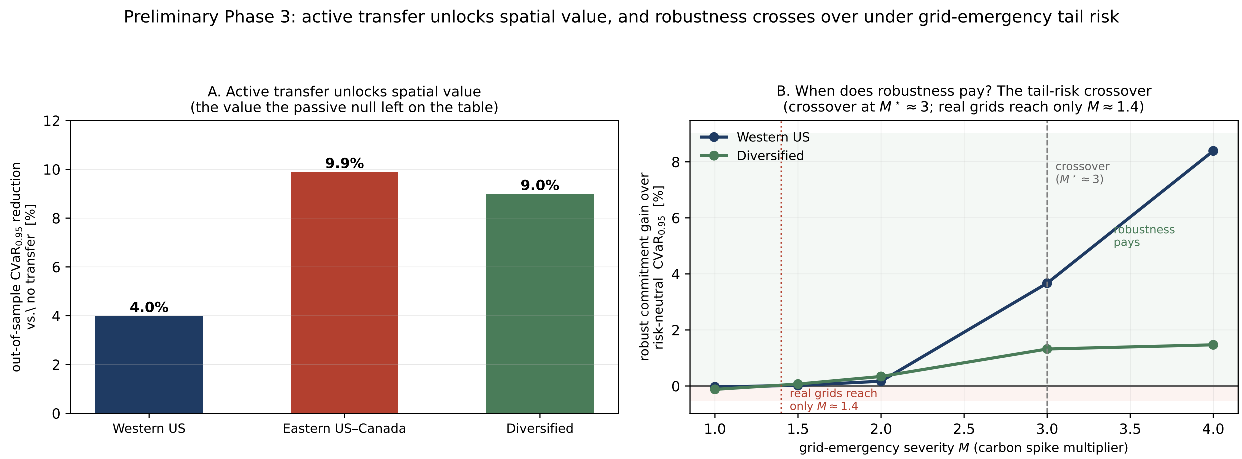

Figure 12: Transfer module validation. **A:** active transfer reduces out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the $\Phi = 0$ baseline. **B:** the severity crossover, the robust (CVaR-hedged) commitment's advantage over the risk-neutral one as grid-emergency severity M grows; it turns positive only past $M^\star \approx 3$. The same numbers are reported and interpreted in the body (Section 4.9).

The synthesis is a clean phase diagram: distributional robustness is immaterial for carbon-aware scheduling under normal conditions (the mean dominates, whether the spatial structure is modelled passively or acted on actively), but becomes valuable precisely when rare tail events make the mean forecast unreliable. The value of robustness is thus a function of tail risk, the precise characterisation of which, together with a calibrated emergency model and a pre-registered evaluation, is the agenda for the follow-on work.

Implementation and validation. These results are not throwaway script output: the transfer model is implemented as a standalone, unit-tested module (`src/models/transfer_dro.py`), where the one-shot transfer DRO, the two-stage robust commitment, and the costly-recourse evaluation are each exercised by tests pinning the model's invariants, conservation of work under inter-region flows ($\sum_{r,t} y_{r,t} = \sum_r W_r$), monotone emission reduction as the transfer budget grows, and exact recovery of the no-transfer schedule at zero budget. The flow variables $f_{r\to s,t}$ carry no self-loops and respect a transfer-capacity budget, and the two-stage program is solved as an LP per scenario with a shared CVaR epigraph. What remains *preliminary* is therefore the experimental protocol, the stylised emergency model, the modest scenario counts, and the absence of pre-registration, not the optimization itself: the crossover of Figure 12 rests on validated code. This is what makes promoting Part 3 to a rigorous study a matter of protocol and calibration, the steps below, rather than of rebuilding the machinery.

The extension beyond this capstone. These preliminary results are the opening of a planned five-part programme that continues well past the capstone. The capstone is Parts 1–2 (the covariance and copula nulls plus the mean-dominance bound). **Part 3**, sketched above, is the spatially-coupled transfer DRO, the active-decision algorithm and the tail-risk crossover, promoted to a rigorous, pre-registered study with calibrated emergencies and, ideally, affine decision rules for the recourse policy. **Part 4** takes the transfer model *online*: a multistage / rolling-horizon robust controller in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a *learned* forecast (learning-augmented DRO) and evaluated closed-loop on real operational traces, the deployable systems contribution. **Part 5** generalises the mean-dominance bound itself into a problem-class condition for *when* cross-coordinate dependence can ever improve a robust allocation, of which carbon-aware scheduling is one instance and energy storage, EV fleets and spatial supply chains are others, the theory that turns the sequence into a research programme. The programme is detailed in the project's research roadmap; this appendix is only its first, deliberately preliminary, datapoint.

Annex A: Individual Contribution Statement

This Research Capstone was completed individually by **Marco Ortiz Togashi** under the supervision of Prof. Bissan Ghaddar. As sole author, I am responsible for all components of the work:

- **Research design.** Formulated the three research questions, designed the shuffled-marginals falsification test, and specified the pre-registration protocol that separates assumption validity from value.
- **Modelling.** Specified the Mahalanobis–Wasserstein DRO scheduling model and its SOCP reformulation, the constraint regimes (including the variable carbon-coupled capacity constraint adapted from the SoCC'25 formulation), and the mean-ablation and tail-dependence diagnostics.
- **Data and implementation.** Assembled the multi-grid carbon-intensity and temperature panels, implemented the full experimental pipeline, the robustness battery, the Benjamini–Hochberg correction, and the figures, under version control with a continuous-integration test suite.
- **Analysis and writing.** Produced and interpreted all results, established the causal mechanism, and wrote the manuscript in full.

Generative AI tools were used as a coding and drafting assistant under my direction, as declared on the cover page; all research questions, modelling decisions, experimental designs, interpretations, and final text are my own. The inter-region transfer channel is developed under the supervisor's direction and is identified throughout as the author's day-ahead migration scheduler.

Marco Ortiz Togashi | Date